SCHWARZ RESEARCH INTELLIGENCE

# Web Interface User Journeys

Comprehensive End-to-End Workflows, Interaction Topologies & Production Screen Specifications

| | |
|---|---|
| TARGET PLATFORM<br>https://research-scanner.schwarz | SYSTEM DEPLOYMENT<br>Production (Active) |
| PUBLICATION DATE<br>9 October 2026 | SCOPE<br>Journeys 01–07 & Architecture Appendices |

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# Schwarz Research Intelligence (SRI) — Web Interface User Journeys

Welcome to the comprehensive user journey documentation for the **Schwarz Research Intelligence (SRI)** web platform. This repository section documents the complete end-to-end user workflows, interactive screen topologies, and operational interfaces captured directly from the live production deployment (`https://research-scanner.schwarz` / `https://sri-prod.apps.01.cf.eu01.onstackit.cloud`).

## 🗺️ User Journey Documentation Index

The user journeys are documented in dedicated, modular specifications corresponding to each primary application view:

| # | User Journey | Route | Key Features & Focus | Documentation |
|---|---|---|---|---|
| 01 | **Authentication & Login** | `/` or `/login` | Credentialed entry, token refresh, security notices, radar scan backdrop | 📖 01_login.md |
| 02 | **Executive Dashboard** | `/dashboard` | Recent briefings ledger, multi-run preservation, ad-hoc briefing trigger | 📖 02_dashboard.md |
| 03 | **Briefings Archive** | `/briefing` | Domain filter chips, multi-run history, screening yields, edition previews | 📖 03_briefings.md |
| 04 | **Selected Document** | `/papers/[id]` | Paper metadata, TUM ecosystem/GDPR compliance badges, scorecard, radar chart | 📖 04_selected_document.md |
| 05 | **Pipeline Runs** | `/runs` | Global KPIs (spend, volume, pass rate), status tabs, complete run audit ledger | 📖 05_pipeline_runs.md |
| 06 | **Run Details** | `/runs/[id]` | Deep-dive telemetry, token counts, dollar costs, scraper yield & DAG stages | 📖 06_run_details.md |
| 07 | **Ask Document** | `/papers/[id]` | Full-text grounded AI interrogation drawer, visual citations, live streaming | 📖 07_ask_document.md |
| 08 | **Paper Comments & Collaboration** | `/papers/[id]` | Threaded message feed, Markdown annotations, consensus emoji reactions (👍, 💡, 🚀) | 📖 08_paper_comments.md |
| 09 | **All Papers Corpus** | `/papers` | Cross-domain literature repository, multi-faceted filtering, sorting, pagination | 📖 09_all_papers.md |

## 📸 Production Screenshot Gallery

All screenshots were captured against the live production environment using automated Playwright tests ( `tests_e2e/capture_journeys.mjs` ) authenticated with the production administrative

credentials:

1. **Login Page**: `./screenshots/01_login.png`
2. **Dashboard**: `./screenshots/02_dashboard.png`
3. **Briefings Archive**: `./screenshots/03_briefings.png`
4. **Selected Document (Sequential Scroll Steps & Full-Page)**:
   - Step 1 (Masthead, Badges, Abstract, Scorecard & Radar): `./screenshots/04_selected_document_part1.png`
   - Step 2 (Assessment Kickoff & Research Metric Facets): `./screenshots/04_selected_document_part2.png`
   - Step 3 (Schwarz Strategic Fit & Synergies): `./screenshots/04_selected_document_part3.png`
   - Step 4 (Strategic Need Match, TRL Classification & Provenance Footer): `./screenshots/04_selected_document_part4.png`
   - Complete Full-Page Overview: `./screenshots/04_selected_document_fullpage.png`
5. **Pipeline Runs Overview**: `./screenshots/05_pipeline_runs.png`
6. **Pipeline Run Details (DAG)**:
   - Executive Telemetry & Top Stages: `./screenshots/06_run_details.png`
   - Complete 9-Stage DAG Breakdown: `./screenshots/06_run_details_fullpage.png`
7. **Ask Document Interrogation Drawer**: `./screenshots/07_ask_document.png`
8. **Paper Comments & Annotator Collaboration**: `./screenshots/08_paper_comments.png`
9. **All Papers Directory / Corpus Catalog**:
   - Executive Viewport: `./screenshots/09_all_papers.png`
   - Full-Page Catalog: `./screenshots/09_all_papers_fullpage.png`

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## 👥 Target Personas & Primary Objectives

- **Executive & Research Lead (Manager)**:
  - Quickly reviews morning briefings on `/dashboard` and `/briefing`.
  - Audits pipeline operational spend and screening selectivity on `/runs`.
  - Triggers ad-hoc domain scans via the **New Briefing** panel.
- **Intelligence Analyst & Researcher (Analyst)**:
  - Drills into high-signal scientific literature on `/papers/[id]`.
  - Evaluates technology readiness (TRL 1–9) and strategic Schwarz ecosystem fit.
  - Converses with complex papers using the **Ask Document** interactive assistant.
- **System Administrator / Operations**:

- Inspects scraper health (arXiv, OpenAlex), deduplication efficiency, and token burn on `/runs/[id]`.

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## 🔗 Related Architectural Specifications

- 👥 **personas.md** — User Personas, Daily Workflows & Permissions Matrix.
- 🧭 **route_matrix.md** — URL routes, parameter bindings, and backend API contracts.
- 📐 **wireframes.md** — Full-fidelity layout wireframes and screen topologies.
- 🗺️ **user_journeys.md** — Interaction Architecture & State Transitions.

# User Journey: Authentication & Workspace Login

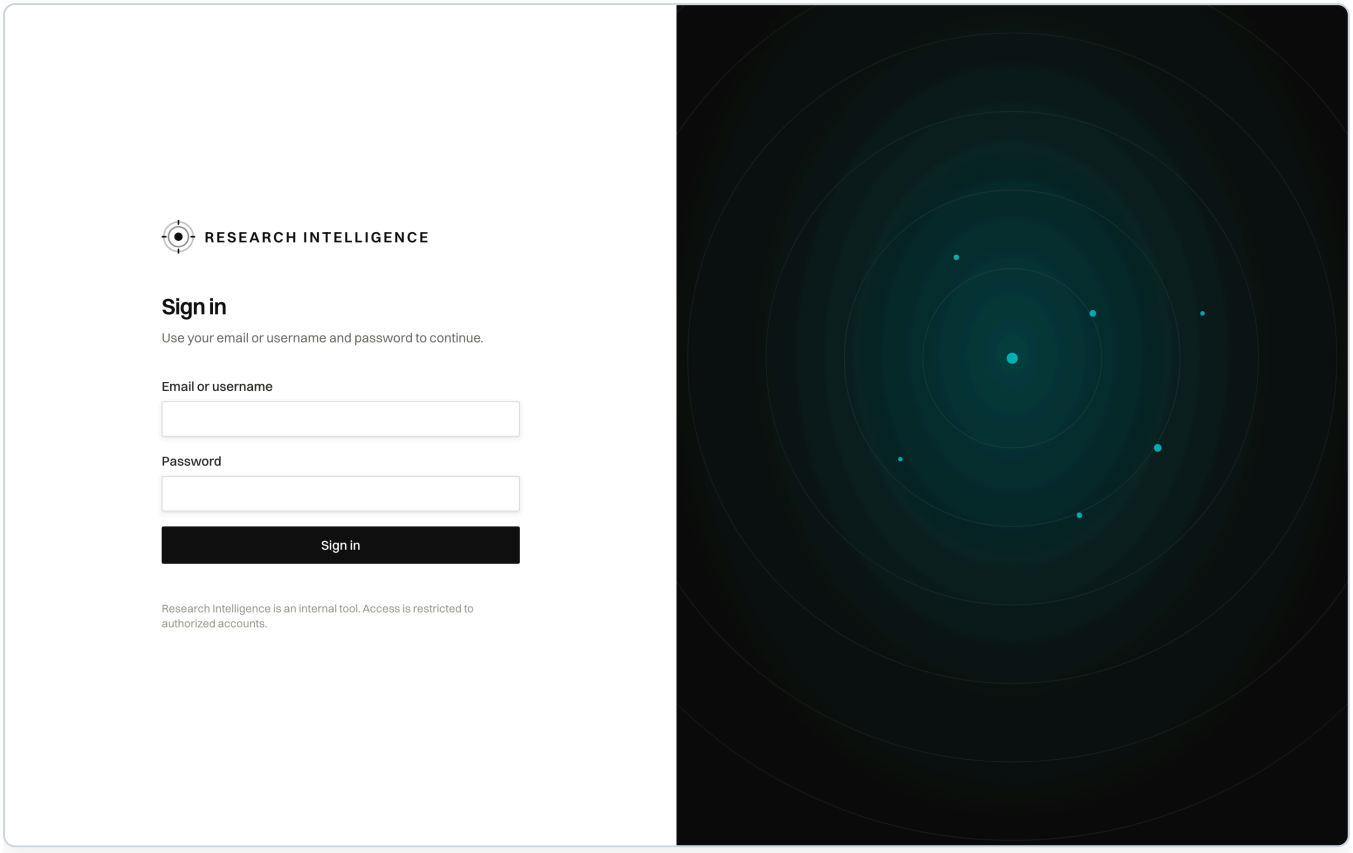
## 1. Overview & Objective

The **Authentication & Login Journey** represents the gateway for authorized personnel (System Administrators, Research Leads, and Intelligence Analysts) accessing the **Schwarz Research Intelligence (SRI)** platform.

The objective of this journey is to provide secure, credentialed access to domain-scoped research intelligence, establish authenticated sessions with automated token refresh, and direct the user to their executive dashboard.

## 2. Interface View

The login interface features the branded Schwarz Research Intelligence radar backdrop and clean credential entry:




## 3. Navigation & URL Route

- **Primary Route**: `/` or `/login`

- **Authenticated Redirection**: If a user already possesses active session cookies ( `sri_access` , `sri_refresh` ), the route automatically redirects to `/dashboard` .
- **Post-Authentication Destination**: `/dashboard` (or the URL specified in the `?next=` query parameter).
- **Session Timeout Handling**: If a session expires during active use, the user is redirected to `/?reason=expired` with an informational notice: *"Your session expired. Please sign in again."*
- **Explicit Sign-out**: Users logging out are redirected to `/?reason=signed-out` with confirmation: *"You have been signed out."*

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## 4. UI Components & Screen Topology

The login screen is split into two visual regions:

1. **Authentication Form Panel (Left)**:
   - **Branding Header**: SRI reticle icon and title `RESEARCH INTELLIGENCE` .
   - **Form Heading**: `Sign in` with descriptive subtext: *"Use your email or username and password to continue."*
   - **Identifier Input ( `#login-identifier` )**: Accepts corporate email address or system username (e.g. `admin` ).
   - **Password Input ( `#login-password` )**: Masked secret input supporting standard security guidelines.
   - **Submit Action**: High-contrast black CTA button `Sign in` .
   - **Access Notice**: *"Research Intelligence is an internal tool. Access is restricted to authorized accounts."*
2. **Branded Scanner Field (Right)**:
   - Faint concentric circular radar sweep rendered in SVG.
   - Active radar markers plotted in teal ( `#00c3cd` ), symbolizing ongoing research scanning and signal detection across global paper repositories.

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## 5. Step-by-Step User Workflow

1. **Accessing the Portal**:
   - Navigate to the production URL: `https://research-scanner.schwarz/` *(or `https://sri-prod.apps.01.cf.eu01.onstackit.cloud/` )*.
   - If unauthenticated, the application displays the `/login` view.
2. **Entering Credentials**:
   - Input username or email (e.g., `admin` ).
   - Input the assigned password (e.g., `schwarz123!` ).

3. **Submission & Authentication**:
   - Click `Sign in` or press `Enter`.
   - The frontend issues a `POST /auth/login` request to the backend auth service.
   - Upon successful verification, secure HTTP-only cookies are established:
     - `sri_access`: Short-lived JSON Web Token for API queries.
     - `sri_refresh`: Secure token enabling seamless background session rotation.
     - `sri_user`: User metadata payload (name, email, roles).
4. **Landing on Dashboard**:
   - The browser transitions smoothly to `/dashboard`, greeted by the personalized executive briefing ledger.

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## 6. Edge Cases & Security Guarantees

| Scenario | System Behavior | User Guidance |
|---|---|---|
| **Invalid Credentials** | Returns HTTP 401 Unauthorized; displays inline error *"Invalid email/username or password."* | Re-check typing or contact the system administrator to reset credentials. |
| **Expired Session** | Proxy detects expired access token and failed refresh, redirecting to `/?reason=expired`. | User enters credentials again to re-acquire fresh tokens. |
| **Restricted Network** | Protected behind corporate CORS & trusted origins (`research-scanner.schwarz`, STACKIT cloud). | Connections from unlisted origins receive HTTP 403 Forbidden. |
| **CSRF Protection** | All mutation endpoints enforce standard CSRF protection matching `CSRF_TRUSTED_ORIGINS`. | Prevents cross-site request forgery attacks during authentication. |

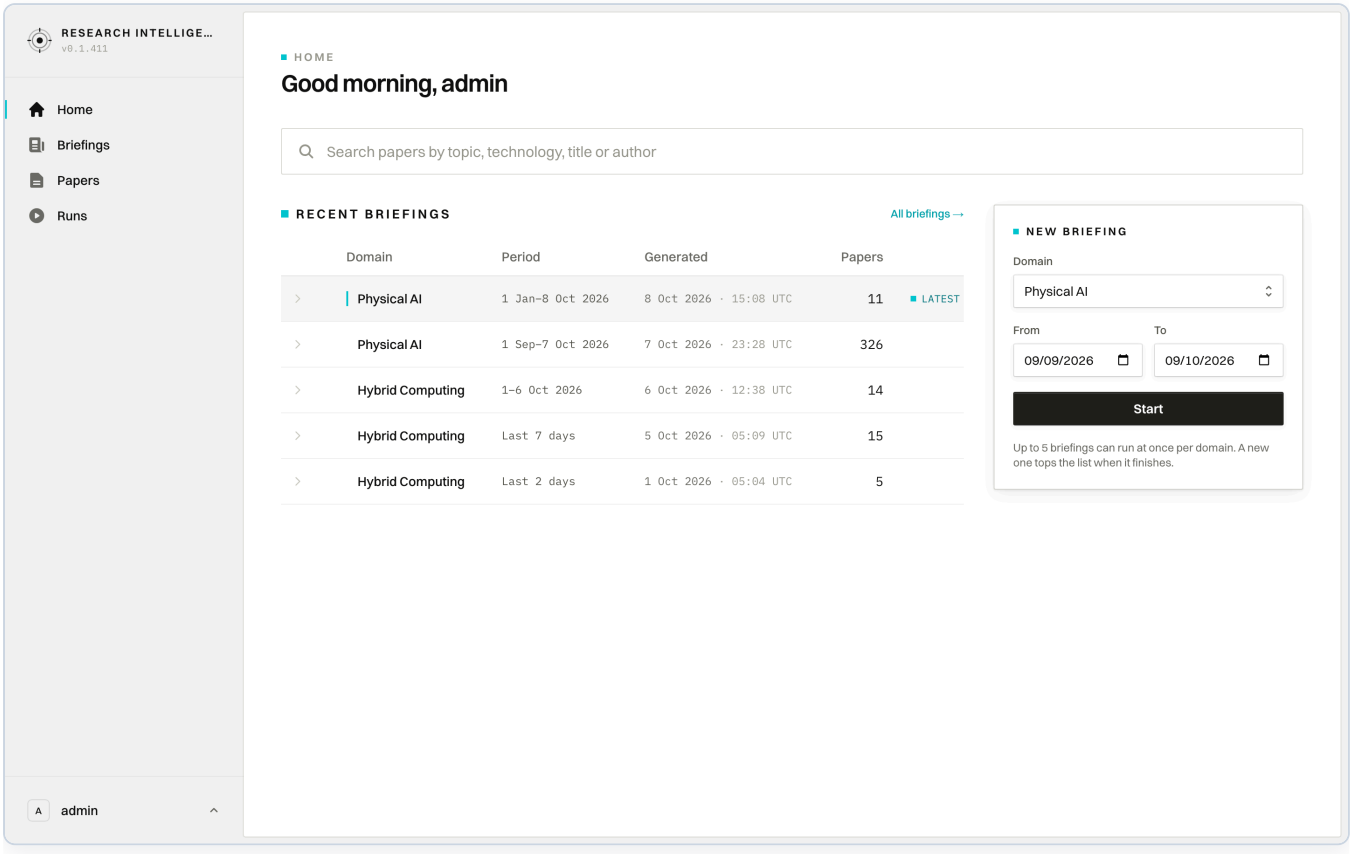
# User Journey: Executive Dashboard & Workspace Hub

## 1. Overview & Objective

The **Dashboard (Home)** is the primary command center for Research Leads and Analysts. It provides an immediate executive summary of recent intelligence generation across all configured domains, allows quick inspection of recently screened papers, and offers a frictionless trigger to initiate new research briefings.

## 2. Interface View

The dashboard combines a chronological ledger of recent briefings with an on-demand briefing execution card:




## 3. Navigation & URL Route

- **Primary Route**: `/dashboard`
- **Sidebar Position**: `Home` (top item in sidebar navigation)
- **Deep-Links**:

- `All briefings →` navigates to `/briefing` (Full Archive)
- Clicking any briefing row navigates to its full edition view: `/briefing/[date]?domain=[workspaceId]&run=[runId]`
- Clicking the expand chevron (`>`) unfolds a preview list of top-scoring papers directly inline.

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## 4. UI Components & Screen Topology

The dashboard layout consists of three primary regions:

### 4.1 Global Sidebar Navigation (Left Rail)

- **Application Brand**: `RESEARCH INTELLIGENCE` with current deployment build (`v0.1.298`).
- **Navigation Links**:
  - `Home` (Active tab, highlighted with icon)
  - `Briefings` (Archive and domain-specific ledgers)
  - `Runs` (Pipeline execution monitoring and cost audit)
- **User Profile Capsule (Bottom)**: Displays active user handle (e.g., `admin`) and quick account controls.

### 4.2 Recent Briefings Ledger (Center / Main)

- **Page Eyebrow & Greeting**: `HOME` / `Good morning, [User]` (dynamically resolved from session).
- **Section Heading**: `RECENT BRIEFINGS` accompanied by quick-nav link `All briefings →`.
- **Multi-Run Preservation**: The ledger preserves multiple runs per domain over time (rather than collapsing each domain to only its single latest run). Analysts can review historical runs side-by-side:
  - **Expand Chevron (`>`)**: Toggles inline view of top-rated papers with their title and scoring snippet.
  - **Domain**: Workspace identifier (e.g., *Physical AI*, *Hybrid Computing*, *Chip Design*). A teal indicator marks the newest active run.
  - **Period**: Target date window analyzed by the pipeline (e.g., `11–17 Aug 2026`, `11 Aug–10 Sep 2026`).
  - **Generated**: Completion timestamp in UTC (e.g., `10 Sep 2026 · 08:40 UTC`).
  - **Papers**: Total count of research papers meeting quality screening criteria.
  - **Status Pill (`LATEST`)**: Highlighted badge identifying the most recently completed intelligence cycle.

### 4.3 Start New Briefing Panel (Right Card)

- **Card Header**: `NEW BRIEFING`.
- **Domain Selector**: Dropdown to select the target workspace domain (e.g., *Physical AI*, *Chip Design*, *Hybrid Computing*).
- **Date Range Pickers**:
  - `From`: Beginning of the search horizon.
  - `To`: End of the search horizon (defaults to the current date).
- **Execution Action**: High-contrast `Start` button.
- **Concurrency & Mutex Note**: *"Briefings run one at a time. When this one finishes, it tops the list."*

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## 5. Step-by-Step User Workflow

### 5.1 Reviewing Recent Intelligence

1. Navigate to `/dashboard` upon logging in.
2. Review the `RECENT BRIEFINGS` table to see which domains have recently refreshed data.
3. Click the chevron (`>`) next to any briefing row to preview its top papers without leaving the dashboard.
4. Click on the briefing row or paper title to drill down into the full briefing edition or individual paper scorecard.

### 5.2 Triggering an Ad-Hoc Briefing Run

1. Locate the **NEW BRIEFING** card on the right side of the dashboard.
2. In the **Domain** dropdown, choose the target research workspace.
3. Specify the **From** and **To** publication date bounds.
4. Click **Start**.
5. The system initiates an asynchronous pipeline run. When the run finishes, the newly generated briefing automatically tops the `RECENT BRIEFINGS` list.

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## 6. Key Invariants & Behavioral Rules

- **Non-Collapsing Multi-Run Display**: Previous editions for the same domain remain accessible in the chronological ledger, ensuring research teams can track domain evolution over successive runs.
- **Graceful Loading & Suspense**: If the backend is compiling metrics, skeleton placeholders seamlessly animate into real data via React `<ViewTransition>`.

- **Single-Flight Concurrency**: To prevent resource exhaustion, only one pipeline run executes at a time per workspace instance.

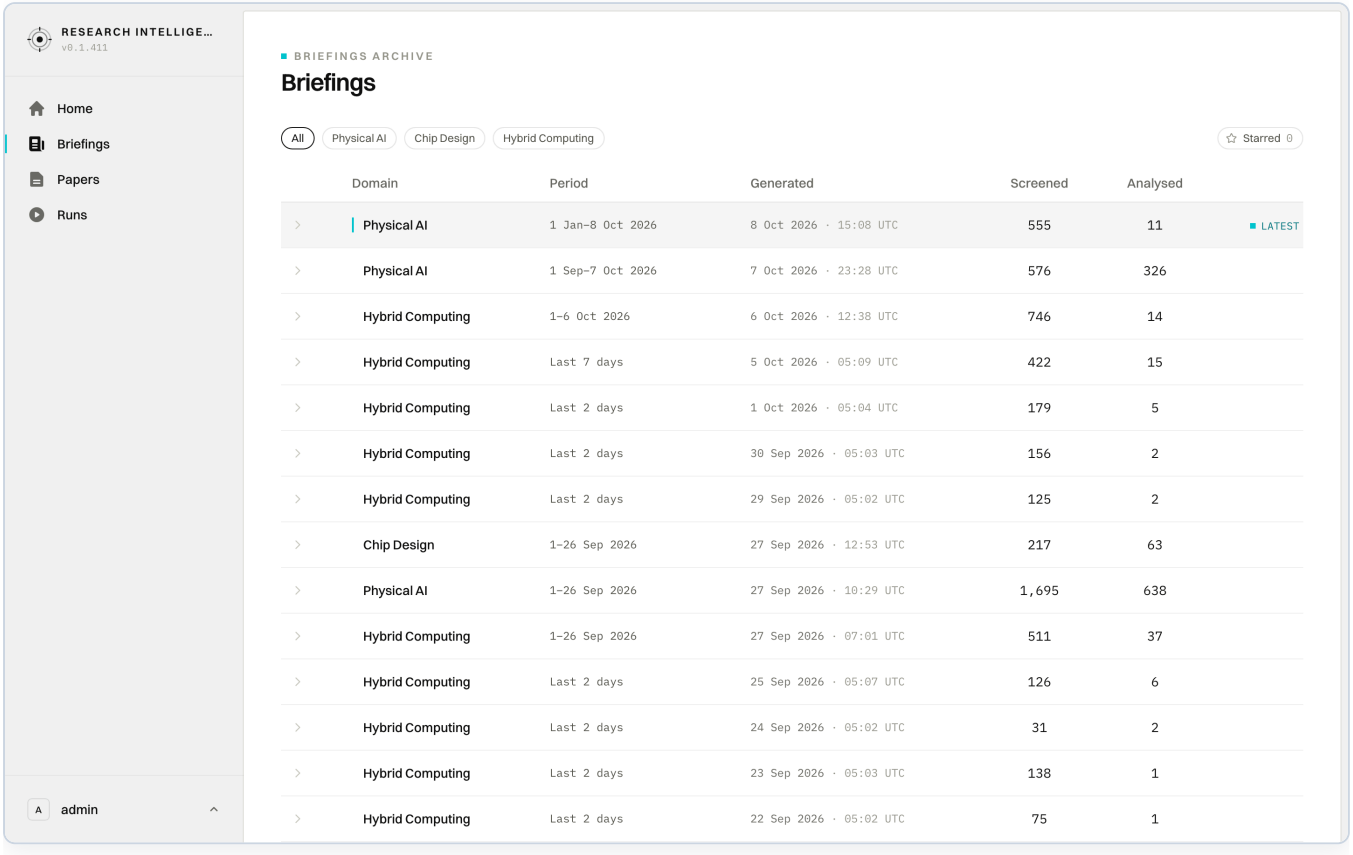
# User Journey: Research Briefings & Multi-Run Domain Intelligence

## 1. Overview & Objective

The **Briefings Archive** ( `/briefing` ) is the comprehensive intelligence repository for all completed research syntheses across every domain. It serves as the primary research reading room, enabling analysts and executives to filter briefings by scientific domain, review pipeline screening conversion rates, track historical editions, and bookmark vital research runs.

## 2. Interface View

The Briefings archive presents an interactive ledger with domain-specific filter chips and screening metrics:




## 3. Navigation & URL Route

- **Primary Route**: `/briefing`
- **Filter Parameters**:

  - `?domain=[domain_slug]` (e.g., `?domain=physical-ai` or `?domain=hybrid-computing`)
  - `?starred=true` (shows only bookmarked briefings)
  - `?page=[N]` (paginated browsing for extensive historical runs)

- **Edition Permalinks**:
  - Clicking any row navigates to the detailed edition: `/briefing/[date]?domain=[workspaceId]&run=[runId]`

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## 4. UI Components & Screen Topology

### 4.1 Filter Strip & Quick Controls

- **Domain Filter Chips**:
  - `All`: Aggregated view across all configured workspaces.
  - Domain-specific chips (e.g., `Physical AI`, `Chip Design`, `Hybrid Computing`): Clicking a chip filters the ledger instantly to that single domain without requiring a full page reload.

- **Starred Filter Button (`☆ Starred [count]`)**: Toggles view to only show user-bookmarked briefing editions for quick retrieval during executive reviews.

### 4.2 Comprehensive Multi-Run Ledger

The ledger preserves full historical fidelity—every run executed for each domain is preserved as an individual row rather than collapsed into a single summary.

| Column | Description | Example |
|---|---|---|
| `>` **(Expand Chevron)** | Unfolds an inline summary of the top papers included in the briefing edition. | Inline title preview & score snippet |
| **`Domain`** | Research domain workspace name. Active/latest run displays a vertical teal indicator bar. | `Physical AI`, `Hybrid Computing`, `Chip Design` |
| **`Period`** | Publication date window scanned by the pipeline. | `11–17 Aug 2026`, `11 Aug–10 Sep 2026` |
| **`Generated`** | UTC completion timestamp. | `10 Sep 2026 · 08:40 UTC` |
| **`Screened`** | Total number of candidate papers fetched from upstream scholarly databases (arXiv, OpenAlex). | `416`, `178`, `60`, `7` |
| **`Included`** | Final number of papers that successfully passed rigorous LLM abstract and full-text screening. | `34`, `19`, `16`, `1` |
| **`LATEST` Badge** | Highlighted status pill affixed to the most recent run in the domain. | Teal indicator badge |

## 5. Step-by-Step User Workflow

### 5.1 Filtering Intelligence by Domain

1. Open `/briefing` from the left sidebar.
2. The initial view defaults to **`All`**, showing a chronological stream of runs across all domains.
3. Click a domain chip (e.g., **`Physical AI`** or **`Hybrid Computing`**) to isolate research runs for that specific topic.
4. The URL updates automatically (e.g., `/briefing?domain=physical-ai`), creating a shareable permalink.

### 5.2 Evaluating Screening Selectivity

1. Inspect the **`Screened`** vs. **`Included`** columns.
   - For example, in a run where **416 papers were screened** and **34 were included**, the pipeline achieved an ~8% acceptance rate, filtering out 92% of off-topic or low-relevance research.
2. Identify anomalous runs or high-yield periods based on inclusion ratios.

### 5.3 Unfolding & Opening a Briefing Edition

1. Click the chevron (`>`) on any row to preview top papers directly in place.
2. Click the briefing row to open the complete edition view (`/briefing/[date]?domain=...&run=...`).
3. From the edition view, click any highlighted paper to open its comprehensive assessment scorecard.

## 6. Key Architectural Highlights

- **Non-Collapsing Multi-Run Architecture**: Early iterations collapsed domain runs to the latest single entry. The current production interface guarantees full historical retention, enabling longitudinal research tracking.
- **Deep-Link State Synchronization**: Filters (`domain`, `starred`, `page`) are serialized directly into the URL search parameters, ensuring that bookmarked or shared URLs always restore the identical filter state.

# User Journey: Document Detail, Assessment & Scorecard

## 1. Overview & Objective

The **Selected Document (Paper Detail) Journey** ( `/papers/[id]` ) provides an exhaustive, multi-layered intelligence assessment for an individual scientific publication. Because deep technical evaluations encompass author affiliations, compliance audits, peer-level abstracts, granular metric facets, and Technology Readiness Levels (TRL), the page spans several vertical screens.

This document walks through the complete interface sequentially from top to bottom, detailing every section revealed as the user scrolls through the document.

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## 2. Complete Visual Walkthrough (Sequential Scroll Progression)

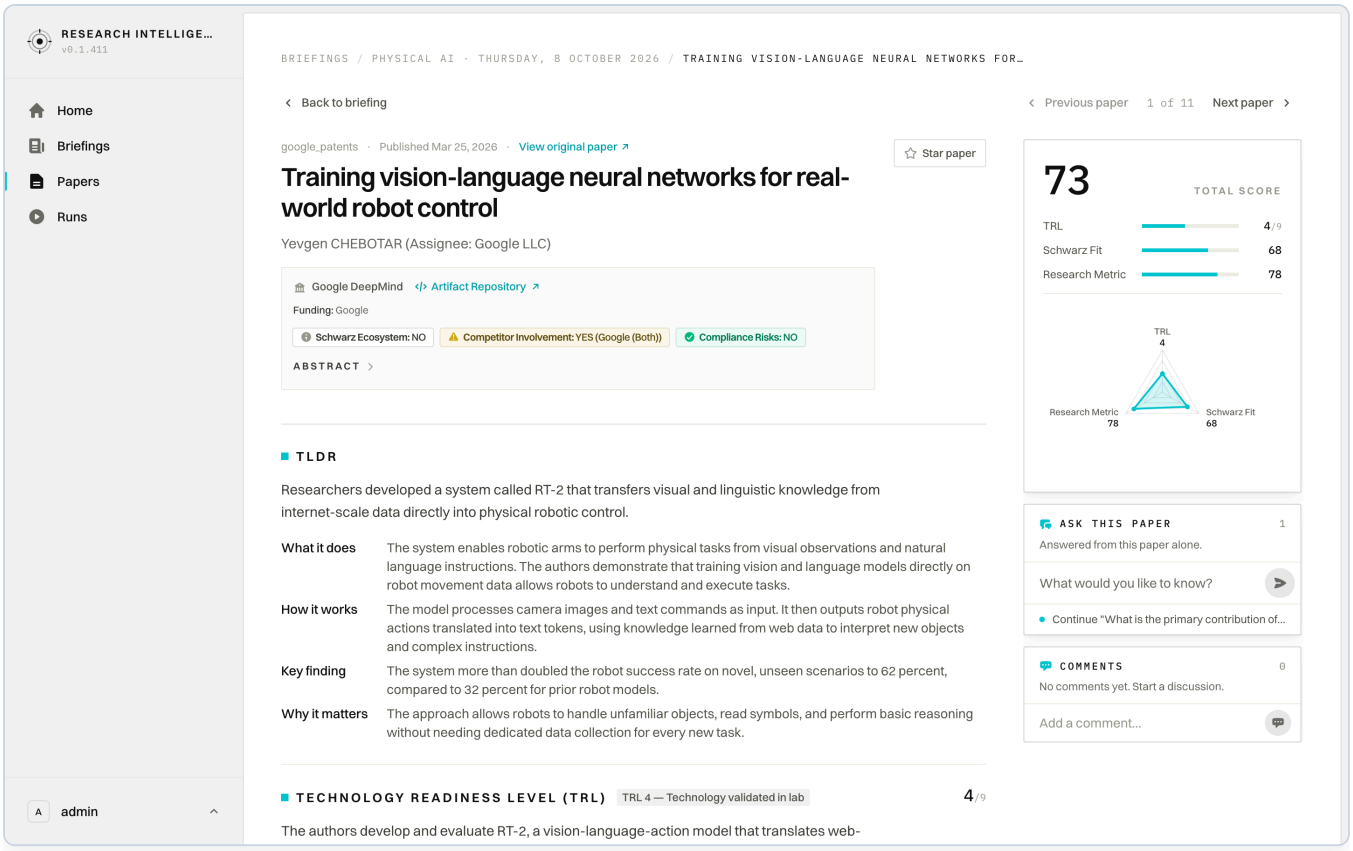
Due to the depth of the LLM extraction framework and technical assessment facets, the document is captured in four cleanly aligned vertical stages from the masthead down to the ingestion provenance footer:

```
[ Step 1: Masthead, TLDR & Scorecard ] → [ Step 2: TRL Classification ] → [ Step 3: Sc
```

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### Step 1: Masthead, Compliance Badges, Abstract & Scorecard Rail

The initial viewport presents the paper's academic provenance, strategic compliance certifications, the scientific abstract toggle, structured TLDR synthesis, and the sticky assessment scorecard:




## Key UI Elements:

- **Top Bar Navigation**: `< Back to briefing` navigates back to the parent briefing edition; `arXiv · Published Sep 29, 2026 · View original paper ↗` offers an outbound link to the upstream preprint repository.
- **Bookmark Trigger**: `☆ Star paper` adds this publication to the analyst's personal shortlist.
- **Publication Masthead**: Title (*"Rho: A Foundation for Efficiently Adaptable VLA Models"*) and author list (*Rho Team, Simran Bagaria, Daphne Chen, Dean Fortier, Jianlong Fu, Michael Harrison, Tess Hellebrekers, Neel Joshi, Andrey Kolobov, Dalton Moore, Galen Mullins, Michael Murray, Eduardo Salinas, Reuben Tan*).
- **Institutional Affiliation & Funding Card**:
  - **Organization**: *Microsoft Research* with direct link to `Artifact Repository ↗`.
  - **Compliance Badges**:
    - `ⓘ Schwarz Ecosystem: NO` (Indicates no prior formal Schwarz partnership).
    - `⚠ Competitor Involvement: YES (Microsoft (Author Affiliation))` (Explicitly alerts leadership to corporate hyperscaler presence).
    - `✓ Compliance Risks: NO` (Confirms absence of IP infringement, sanctions, or regulatory red flags).
- **Abstract & TLDR Synthesis**:
  - **Abstract**: Expandable accordion revealing the unabridged author abstract.
  - **What it does**: *The system introduces an adaptable AI model designed for two-armed robots, allowing robots to quickly learn new manipulation tasks from limited demonstration data and*

*modest human guidance.*

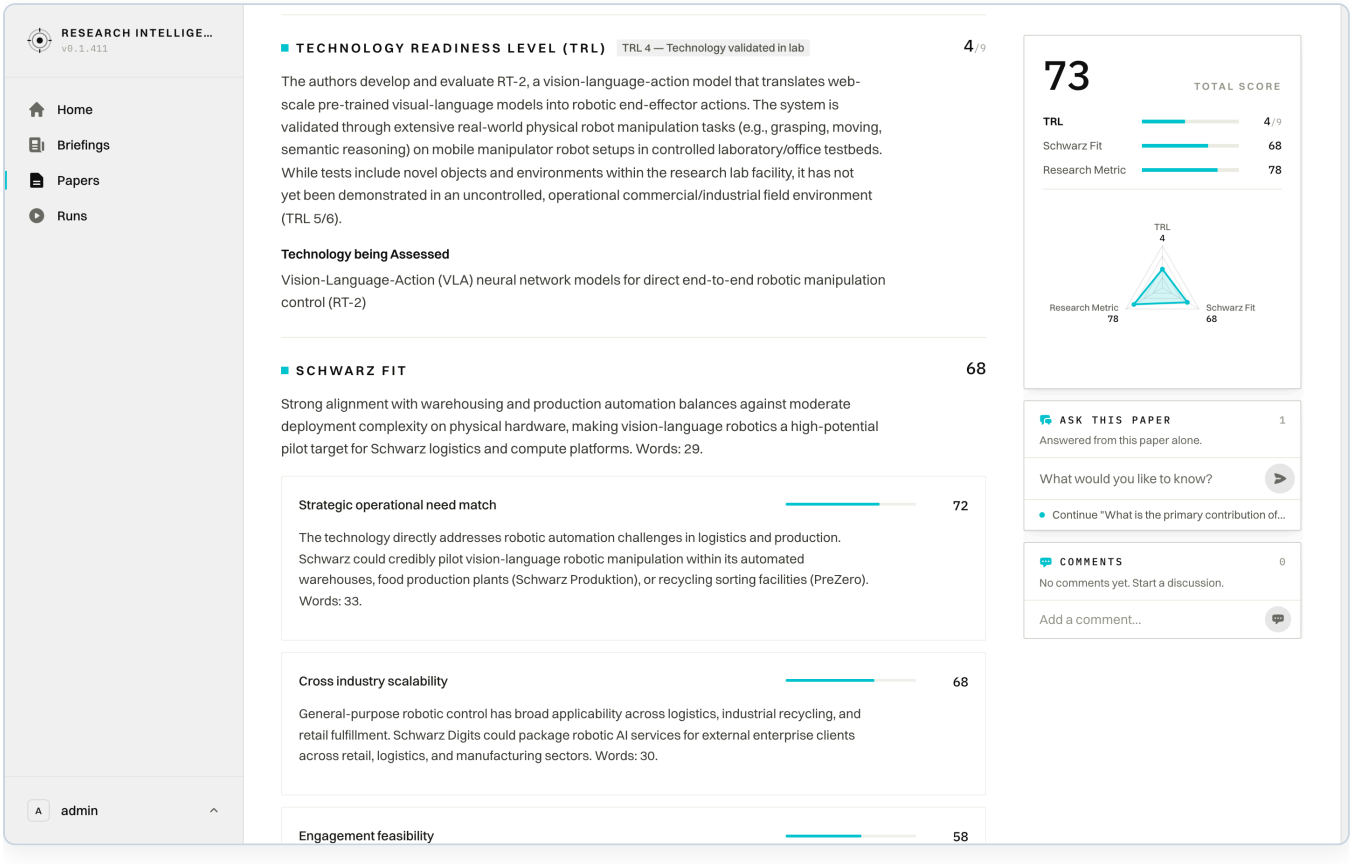
- **How it works**: *The model first learns general physical movements on a specific robot design before tackling specific tasks. It pairs an image-understanding system with a motion generator, updating a small internal steering module during online human interventions.*
- **Key finding**: *Platform-specific midtraining halved the required task demonstration data to hit target success rates and outperformed comparable open-source robot models.*
- **Why it matters**: *Teaching robots new physical behaviors typically demands large sets of custom demonstrations. Splitting hardware familiarization from task training makes robotic deployment faster and less data-intensive.*

- **Sticky Scorecard Rail (Right)**:
  - **Total Score (`73`)**: High composite relevance rating reflecting industrial robotics potential.
  - **Dimensional Score Bars**: `TRL: 4 / 9`, `Schwarz Fit: 70`, `Research Metric: 75`.
  - **Radar Chart**: Triangular visualization balancing lab validation maturity (TRL 4), scientific rigor (75), and strategic corporate fit (70).
  - **Ask This Paper Composer**: Quick question input with trigger (`➤`) opening the interactive chat drawer.

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## Step 2: Technology Readiness Level (TRL) Classification

Scrolling down brings the dedicated **Technology Readiness Level (TRL)** assessment cleanly into view:



Key UI Elements:

- **TRL Classification Badge**: `TRL 4 – Technology validated in lab` (`4 / 9`).
- **Evaluation Evidence**:

> *"The authors developed and integrated a 5.2B parameter VLA model (Rho) featuring a physically grounded Phi-Phy backbone, a flow-matching action expert, embodiment midtraining, and latent-space online adaptation. The integrated system was systematically tested and validated in simulated robotic environments (RoboEval, LIBERO, RobotTwin 2.0, Meta-World) and across multiple real hardware bimanual setups (YAM Box, UR AI Trainer, Franka FR3 Duo) within controlled laboratory conditions. The evaluations do not reflect deployment in a true operational/industrial environment (which would correspond to TRL 6 or 7), making TRL 4 the highest solidly supported level."*

- **Technology being Assessed**:

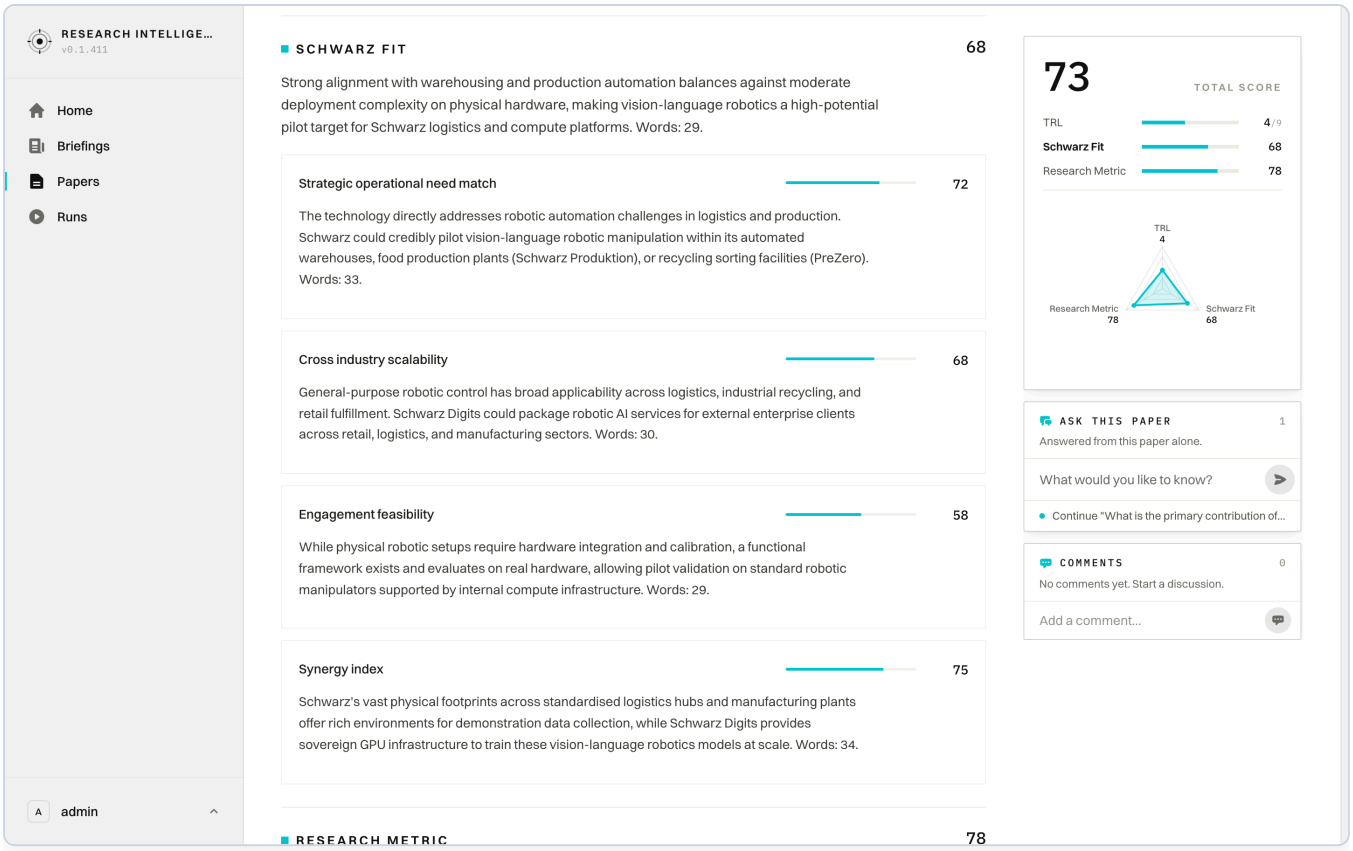
> *"Vision-Language-Action (VLA) foundation model architecture and progressive multi-stage adaptation framework for bimanual robotic manipulation"*

- **Scorecard Rail Persistence**: The right scorecard remains pinned in view, providing unbroken context.

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### Step 3: Schwarz Strategic Fit & Enterprise Synergies

Continuing the scroll centers the **Schwarz Fit** evaluation, detailing how bimanual robotics technology impacts Schwarz Group divisions:




## Key UI Elements:

- **Schwarz Fit Headline (`70 / 100`)**: Strong operational alignment across retail distribution centers, manufacturing, and sovereign cloud infrastructure.
- **Lead Strategic Rationale**:

> *"Fit is strongest in operational synergy and external cloud scalability via STACKIT, while held back slightly by the hardware engineering overhead required to deploy and supervise physical bimanual robots in live environments."*

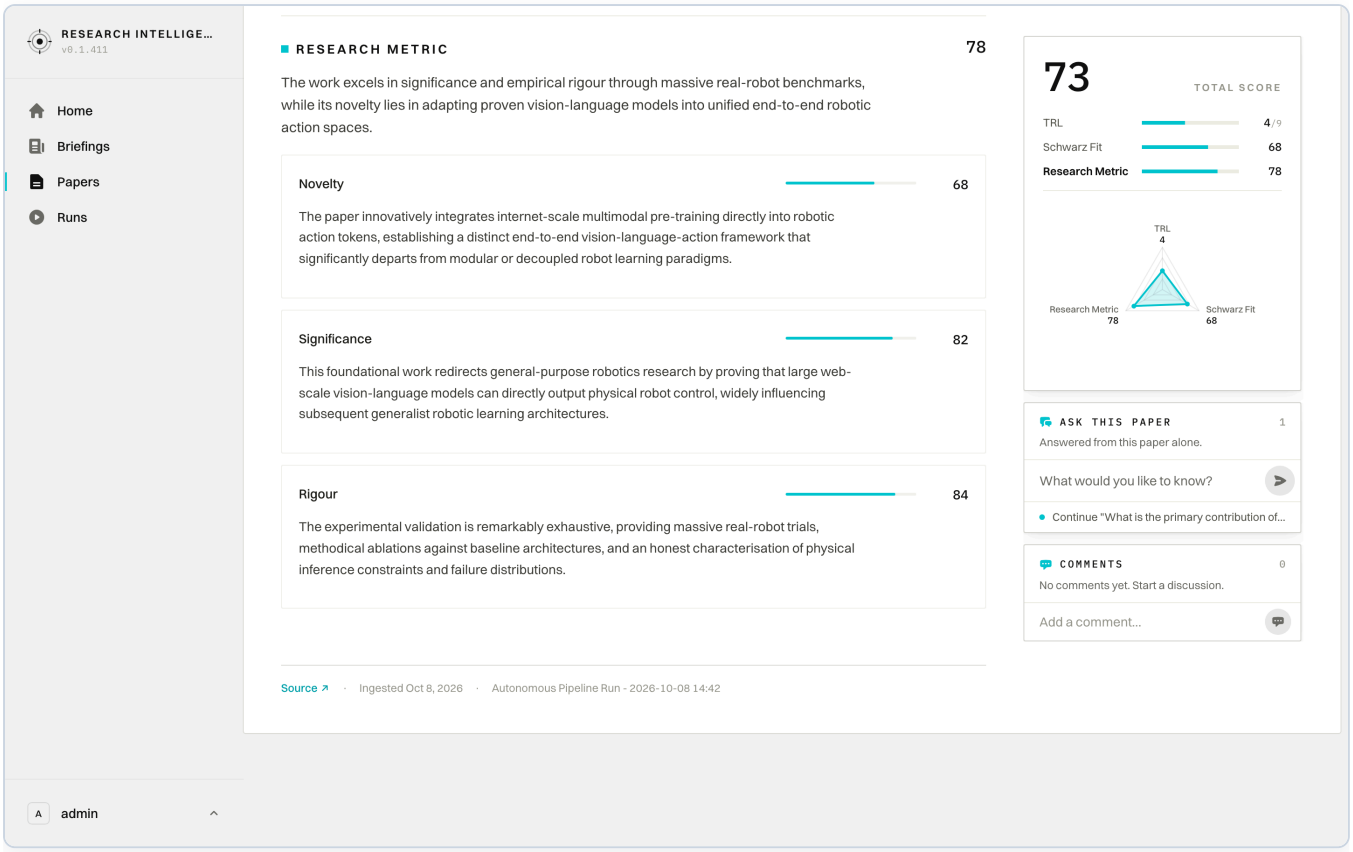
- **Operational Facet Ratings**:
  1. `Strategic operational need match` (`68 / 100`): Rho targets multi-stage industrial manipulation and rapid adaptation, offering Lidl logistics, Schwarz Produktion plants, and PreZero recycling streams a direct path to automating complex packing, sorting, and assembly tasks with fewer robot demonstrations.
  2. `Cross industry scalability` (`75 / 100`): Because Rho is embodiment-agnostic across dual-arm setups and publicly released, Schwarz Digits could productize hosted physical AI adaptation services on STACKIT for external manufacturing, logistics, and retail automation clients across Europe.
  3. `Engagement feasibility` (`65 / 100`): With open weights and codebase already published, Schwarz Digits can pilot models immediately on STACKIT; however, physical deployment still requires dual-arm hardware rigs, teleoperation collection setups, and human-in-the-loop safety oversight.

4. `Synergy index` **(** `72 / 100` **)**: Schwarz possesses massive physical warehouse footprints, standardized logistics cells, and sovereign STACKIT GPU infrastructure, enabling rapid internal training and scaled multi-embodiment deployment far faster than competitors lacking integrated physical operations and owned cloud compute.

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## Step 4: Research Metric & Scientific Provenance

The lower section presents the granular academic rigour decomposition and full ingestion audit trail:




### Key UI Elements:

- **Research Metric Headline (** `75 / 100` **)**: Evaluates scientific novelty, benchmark significance, and experimental methodology.
- **Lead Evaluation Rationale**:

> *"The research excels in rigorous empirical benchmarking and practical significance through open-weight dual-arm checkpoints, while its novelty lies in an effective multi-stage adaptation framing rather than radically new fundamental model architectures."*

- **Granular Facet Scores**:
  1. `Novelty` **(** `62 / 100` **)**: The work innovates by decoupling embodiment midtraining from downstream task specialization in VLAs and integrating flow inversion for latent online

adaptation, though individual components build largely on existing architectures like FlowDagger and DiT-style experts.

2. `Significance` **(** `78 / 100` **)**: By halving required task adaptation data and releasing open-weight foundation and embodiment checkpoints across three major bimanual platforms, this paper delivers high utility and establishes an impactful paradigm likely to guide practical robotic deployment.
3. `Rigour` **(** `85 / 100` **)**: The authors conduct extensive, well-controlled real-world and simulation experiments using parallel A/B testing, systematic architectural ablations, explicit standard deviations, and provide transparent discussion of baseline conditions and hardware limitations.

- **Provenance Audit Strip (Footer)**:
  - **Scholarly Source Link**: `arXiv 2609.38164 ↗` linking directly to the authoritative preprint.
  - **Ingestion Timestamp**: `Ingested Oct 7, 2026`.
  - **Originating Pipeline Run**: Hyperlink to `Autonomous Pipeline Run – 2026-10-07 20:59` allowing auditors to inspect raw scraper outputs and model extraction parameters.

---

## 3. Full-Page Overview

To view the complete, uninterrupted page layout from top to bottom in a single composite graphic:

- 📄 **View Complete Full-Page Screenshot (2896px tall)**

---

## 4. Navigation & URL Route

- **Primary Route**: `/papers/[id]` (e.g., `/papers/1d36d81b-0072-485b-82a9-d7ee548dc363`)
- **Run-Scoped Permalinks**: `/papers/[id]?run=[runId]`
  - Binds the scorecard and facet commentary to the exact pipeline run edition.
- **Parent Navigation**: `< Back to briefing` restores the previous briefing context.

---

## 5. Summary of User Actions on this Page

1. **Scan Metadata & Badges**: Immediately confirm competitor involvement (`Competitor Involvement: YES (Microsoft)`) and compliance status.
2. **Review High-Level Scores**: Consult the sticky sidebar for Total Score (`73`), TRL (`4 / 9`), and Radar balance.
3. **Verify TRL Level**: Read empirical lab justifications validating the model at `TRL 4`.

4. **Audit Strategic Enterprise Fit**: Examine the `Schwarz Fit` facets ( `Synergy: 72` , `Scalability: 75` , `Need Match: 68` , `Feasibility: 65` ) to evaluate deployment across Lidl logistics and STACKIT.
5. **Investigate Academic Rigour**: Read the `Novelty` ( `62` ), `Significance` ( `78` ), and `Rigour` ( `85` ) commentary.
6. **Audit Pipeline Provenance**: Trace the evaluation back to its originating run ( `Autonomous Pipeline Run — 2026-10-07 20:59` ) via the footer link.
7. **Ask In-Depth Questions**: Type into the `ASK THIS PAPER` composer at any point during scrolling to launch the interactive drawer.

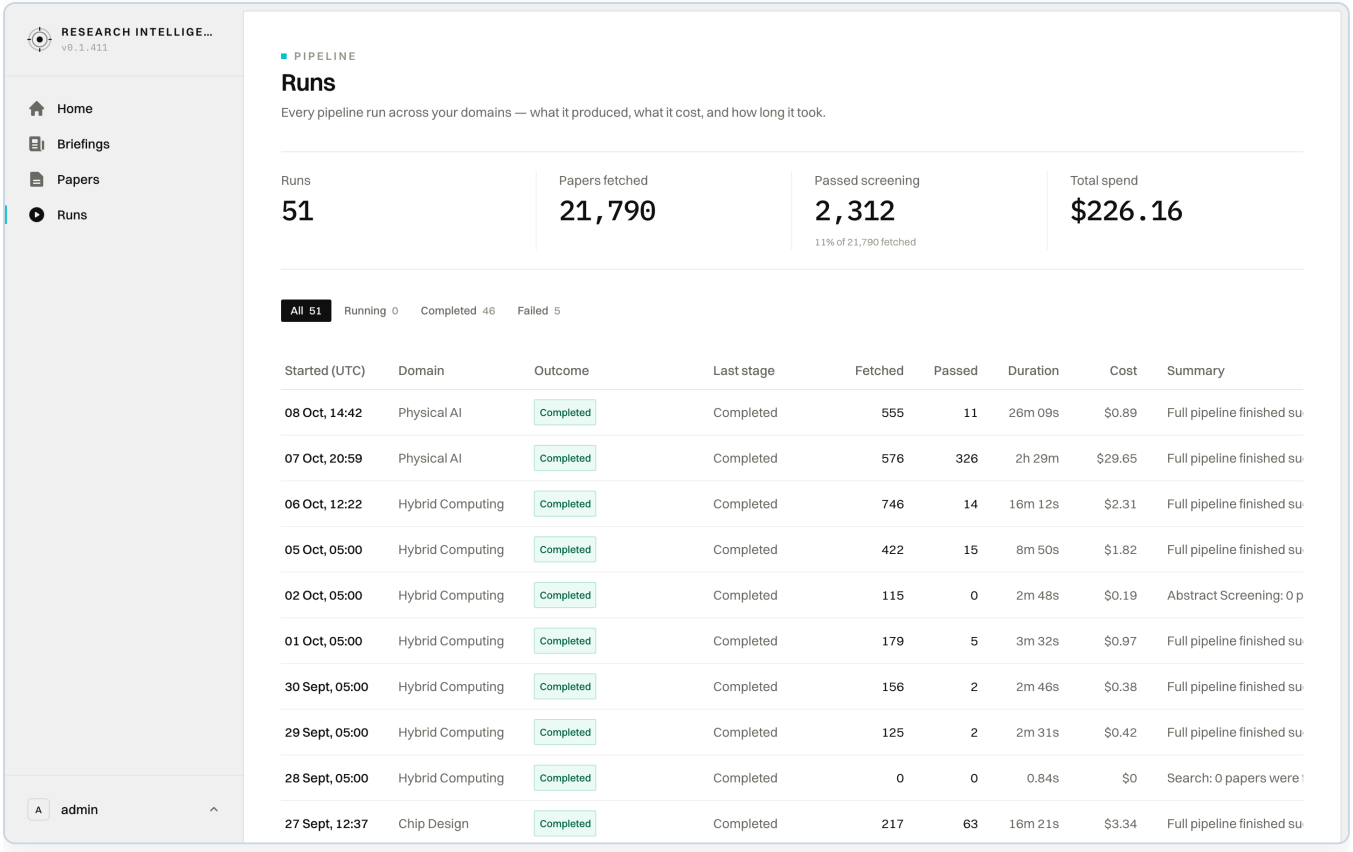
# User Journey: Pipeline Runs Ledger & Execution Monitoring

## 1. Overview & Objective

The **Pipeline Runs Journey** ( `/runs` ) is designed for Operations Managers, Research Leads, and AI Engineers to monitor, audit, and evaluate every automated ingestion and screening pipeline executed across the platform. It provides high-level aggregated operational KPIs (total volume, screening pass rates, cumulative LLM spend) alongside a granular execution log for all historical runs.

## 2. Interface View

The Runs view presents aggregate operational metrics above a filterable execution ledger:




## 3. Navigation & URL Route

- **Primary Route**: `/runs`
- **Sidebar Position**: `Runs` (third item in sidebar navigation)

- **Status Filter Query**:
  - `?status=all` (Default: complete audit trail)
  - `?status=running` (Active live runs)
  - `?status=completed` (Successfully finished runs)
  - `?status=failed` (Faulted or aborted runs)
- **Drill-Down Permalinks**:
  - Clicking any run row navigates to the detailed execution DAG: `/runs/[runId]` (e.g., `/runs/1243db08-42bd-44a3-95fc-f5ee4664c181`)

---

## 4. UI Components & Screen Topology

### 4.1 Executive KPI Metric Cards

A prominent four-cell metric strip summarizes global pipeline performance:

1. **Runs (`19`)**: Total number of automated pipeline runs executed.
2. **Papers fetched (`1,527`)**: Raw scholarly documents scraped from external indexes.
3. **Passed screening (`251`)**: Papers accepted by LLM screening filters (displays overall acceptance efficiency, e.g., `16% of 1,527 fetched`).
4. **Total spend (`$15.33`)**: Cumulative LLM inference cost in USD attributed to scraping, screening, and parameter extractions.

### 4.2 Status Filter Strip

Interactive pill tabs allow instant filtering by run status:

- **`All [count]`** (e.g., `All 19`)
- **`Running [count]`** (e.g., `Running 0`)
- **`Completed [count]`** (e.g., `Completed 19`)
- **`Failed [count]`** (e.g., `Failed 0`)

### 4.3 Detailed Execution Ledger Table

| Column | Description | Example Value |
|---|---|---|
| `Started (UTC)` | Execution start date and time. | `10 Sept, 08:39`, `09 Sept, 21:26` |
| `Domain` | Target research workspace. | `Physical AI`, `Hybrid Computing`, `Chip Design` |
| `Outcome` | Terminal state badge. | Green pill `Completed`, `Running`, `Failed` |
| `Last stage` | Final stage reached by the execution DAG. | `Completed`, `Scraping`, `Screening` |
| `Fetched` | Candidate publications scraped. | `7`, `23`, `60`, `416` |
| `Passed` | Papers passing acceptance criteria. | `1`, `3`, `19`, `34` |
| `Duration` | Wall-clock execution time. | `1m 11s`, `3m 21s`, `12m 53s`, `47.2s` |
| `Cost` | Attributed inference cost ($ USD). | `$0.08`, `$0.28`, `$1.03`, `$2.47` |
| `Summary` | Concise human-readable execution summary. | *"Full pipeline finished successfully. Papers: 7 fetched..."* |

---

## 5. Step-by-Step User Workflow

### 5.1 Auditing Throughput & Budget

1. Navigate to `/runs` via the sidebar.
2. Inspect the **Total spend** and **Passed screening** cards to evaluate platform cost-efficiency.
3. Note the cost-per-paper yield (e.g., ~$0.06 per screened paper).

### 5.2 Filtering & Investigating Historical Runs

1. Click the status tabs (`Running`, `Completed`, `Failed`) to identify anomalies or monitor active background executions.
2. Review the **Outcome** and **Last stage** columns for any partial completions.
3. Click any row to navigate directly to that run's deep-dive execution trace (`/runs/[runId]`).

---

## 6. Operational & Governance Guarantees

- **Transparent Spend Accountability**: Every pipeline run records its exact token consumption and dollar spend, enabling cost attribution by research domain.
- **Audit Immutability**: Historical runs remain immutable once completed. Parameters, paper snapshots, and scoring results cannot be overwritten, providing full reproducibility.

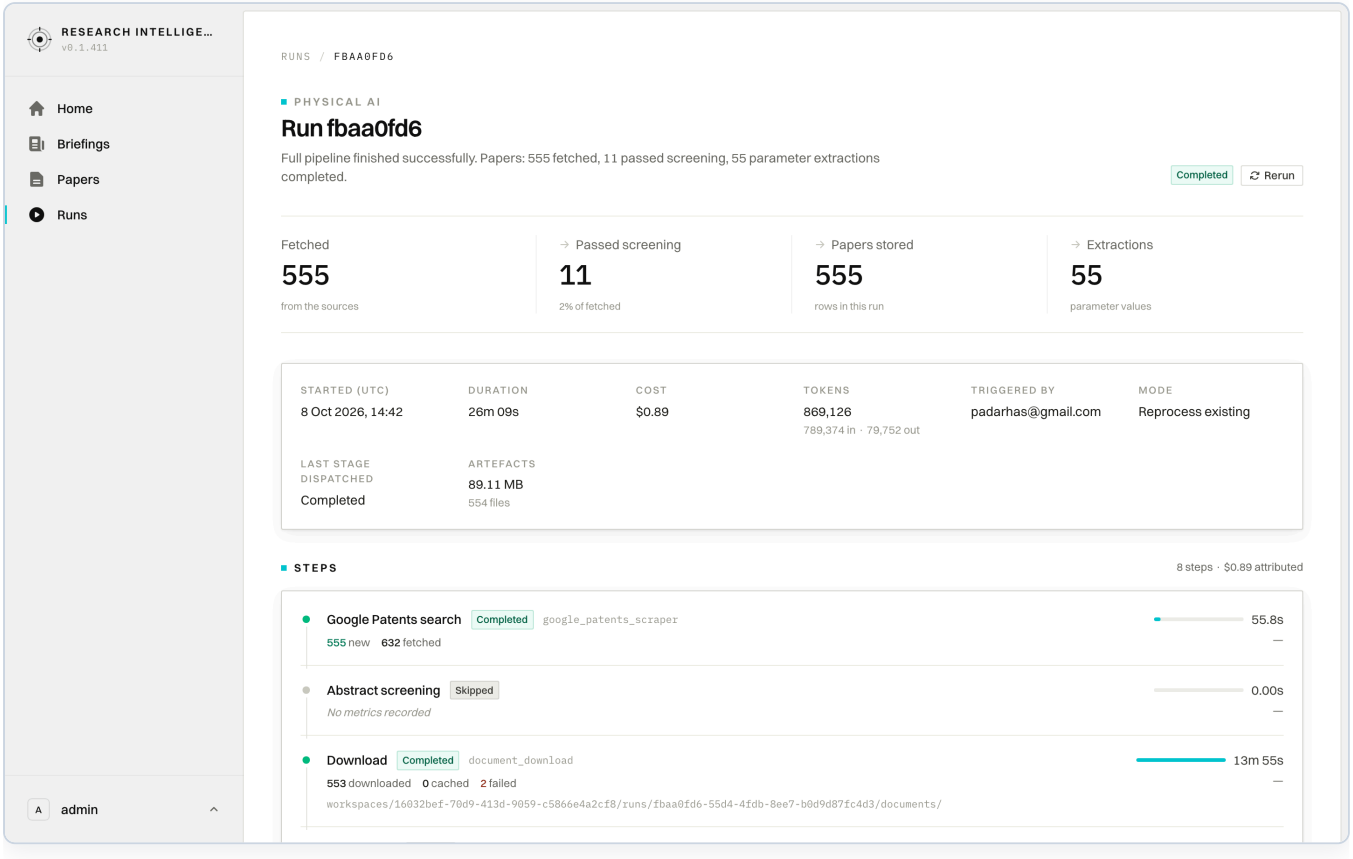
# User Journey: Pipeline Run Deep-Dive & Execution DAG

## 1. Overview & Objective

The **Run Details Journey** ( `/runs/[id]` ) delivers a deep technical trace into a specific pipeline execution. It reveals end-to-end telemetry: ingestion volume, screening conversion rates, token consumption, total inference cost, execution parameters, and a granular step-by-step DAG breakdown detailing upstream scrapers and LLM screening models.

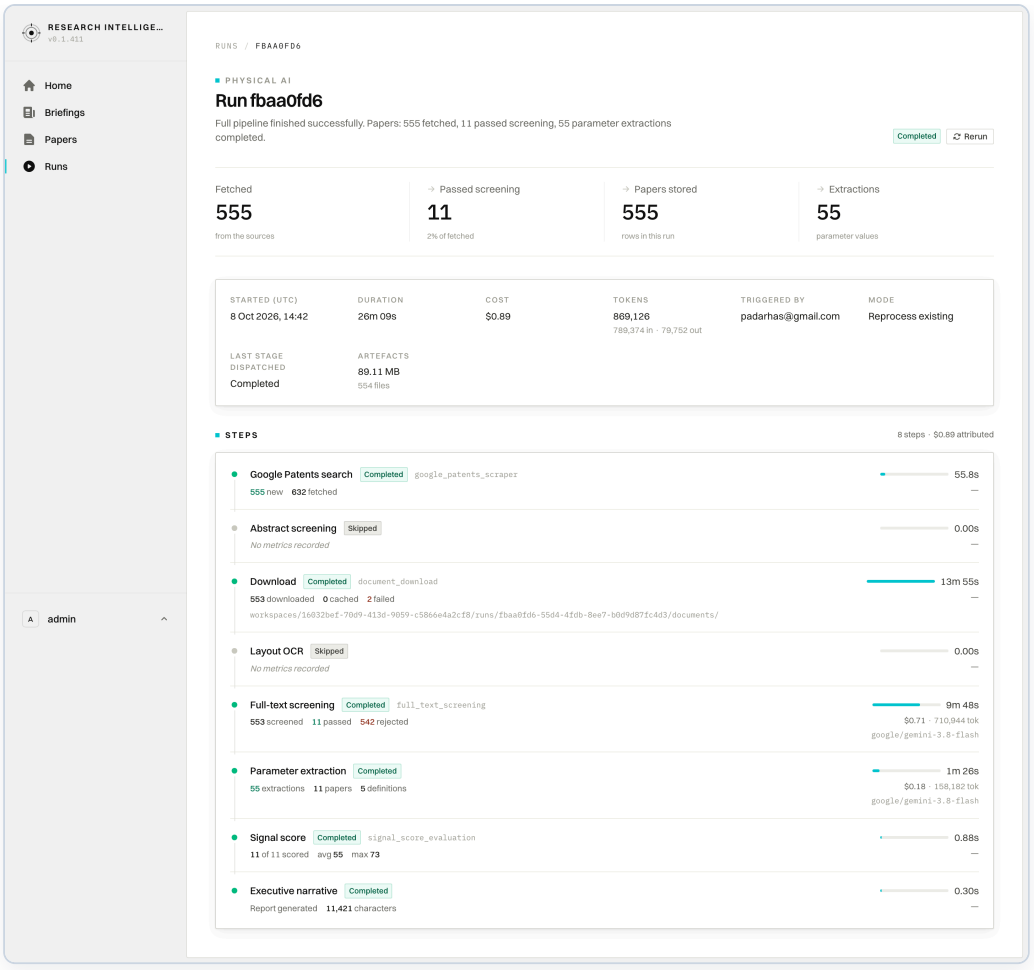
## 2. Interface View

The Run Details view provides a complete post-execution audit log:




### Complete 9-Stage DAG Execution View

The full execution DAG details all upstream scrapers, LLM screening, cached downloads, extraction, and synthesis:




## 3. Navigation & URL Route

- **Primary Route**: `/runs/[id]` (e.g., `/runs/64e8eeec-1c75-4e50-ac95-54a8c99ac175`)
- **Breadcrumb Trail**: `RUNS` (`/runs`) ➔ `[Short UUID]` (e.g., `64E8EEEC`)
- **Access Points**:
  - Clicking any run row in the Runs overview (`/runs`).
  - Accessing the run link from a briefing edition header.

## 4. UI Components & Screen Topology

The view is structured into three primary analytical sections:

### 4.1 Header & Core Metrics Strip

- **Breadcrumbs**: Instant back-link to the full runs ledger.
- **Domain Eyebrow & Run Identifier**: `HYBRID COMPUTING` / `Run 64e8eeec`.
- **Outcome Status Badge**: Green pill `Completed` (or `Running`, `Failed`).

- **Executive Summary**: Comprehensive execution statement (*"Full pipeline finished successfully. Papers: 1058 fetched, 52 passed screening, 265 parameter extractions completed."*).
- **Four Core KPI Cards**:
  - `Fetched` **(1,058)**: Total documents returned by upstream scrapers.
  - `Passed screening` **(52)**: Relevant documents retained (with percentage yield: `5% of fetched`).
  - `Papers stored` **(1,058)**: Persistent bibliographic records archived in this run.
  - `Extractions` **(265)**: Specific scientific and technological parameters extracted.

### 4.2 Run Configuration & Resource Consumption Grid

A dedicated metadata container surfaces operational telemetry:

- `STARTED (UTC)`: Start timestamp (`10 Sept 2026, 21:43`).
- `DURATION`: Total runtime (`22m 38s`).
- `COST`: Exact dollar spend (`$4.94`).
- `TOKENS`: Total LLM token usage (`5,630,345` tokens: `5,391,587 in · 238,758 out`).
- `TRIGGERED BY`: Identity of user or cron trigger (`admin`).
- `MODE`: Processing policy (`Reprocess existing`).
- `LAST STAGE DISPATCHED`: Terminal DAG stage (`Completed`).
- `ARTEFACTS`: Generated intermediate files and PDFs (`136.68 MB · 60 files`).

### 4.3 Step-by-Step DAG Execution Stages

A sequential timeline detailing each discrete pipeline stage (`9 steps · $4.94 attributed`):

1. **arXiv Search (`arxiv_scraper`)**:
   - Status: `Completed`
   - Yield: `518 new`, `519 fetched`
   - Timing: `1m 54s`
2. **OpenAlex Search (`openalex_scraper`)**:
   - Status: `Completed`
   - Yield: `592 new`, `1,000 fetched`
   - Timing: `28.9s`
3. **Abstract Screening**:
   - Status: `Completed`
   - Filtering: `1,058 screened` ➔ `88 passed`, `968 rejected`, `2 withdrawn`
   - Timing & Cost: `9m 59s` | `$1.70 · 1,757,560 tok`
   - Model Attribution: `google/gemini-3.8-flash`

4. **PDF Binary Download (`document_download`)**:
   - Status: `Completed`
   - Yield: **`3 downloaded`** · **`56 cached`** · **`27 failed`** (distinguishes newly downloaded PDFs from warm cache hits)
   - Timing: `1m 00s`
5. **Layout OCR**:
   - Status: `Skipped` (`No metrics recorded`)
6. **Full-Text Screening (`full_text_screening`)**:
   - Status: `Completed`
   - Filtering: `61 screened` ➔ `52 passed`, `9 rejected`
   - Timing & Cost: `1m 50s` | `$0.55 · 675,136 tok`
   - Model Attribution: `google/gemini-3.8-flash`
7. **Parameter Extraction**:
   - Status: `Completed`
   - Yield: `260 extractions`, `52 papers`, `5 definitions`
   - Timing & Cost: `7m 50s` | `$2.69 · 3,197,649 tok`
   - Model Attribution: `google/gemini-3.8-flash`
8. **Signal Score (`signal_score_evaluation`)**:
   - Status: `Completed`
   - Yield: `52 of 52 scored`, `avg 46`, `max 53`
   - Timing: `1.6s`
9. **Executive Narrative**:
   - Status: `Completed`
   - Yield: `Report generated · 19,616 characters`
   - Timing: `0.29s`

---

## 5. Step-by-Step User Workflow

### 5.1 Inspecting Run Health & Throughput

1. From `/runs`, click on a run row to open its details.
2. Review the **Outcome** badge and **Summary** statement.
3. Check the **Fetched** vs. **Passed screening** ratio to verify scraper health and screening selectivity.

### 5.2 Analyzing LLM Spend & Token Efficiency

1. Check the **COST** and **TOKENS** fields in the metadata grid.
2. In the **STEPS** breakdown, locate which stage incurred the highest token usage (typically *Abstract Screening* or *Full-Text Screening*).
3. Verify the underlying model deployed (e.g., `gemini-3.8-flash`) and its cost contribution.

### 5.3 Diagnosing Deduplication Performance

1. Inspect the **DUPLICATES** count in the metadata grid and across individual scrapers (e.g., 134 duplicates caught during arXiv search).
2. Confirm that previously analyzed papers were appropriately deduplicated without incurring redundant LLM inference costs.

---

## 6. Technical Invariants & Troubleshooting

- **Reproducibility**: The exact model checkpoint, prompt versions, and scraper configurations are permanently bound to the run record.
- **Failure Isolation**: If any individual step encounters an error, the DAG halts at that stage, logs the stack trace in the step row, and flags the run as `Failed` with partial artifacts preserved for inspection.

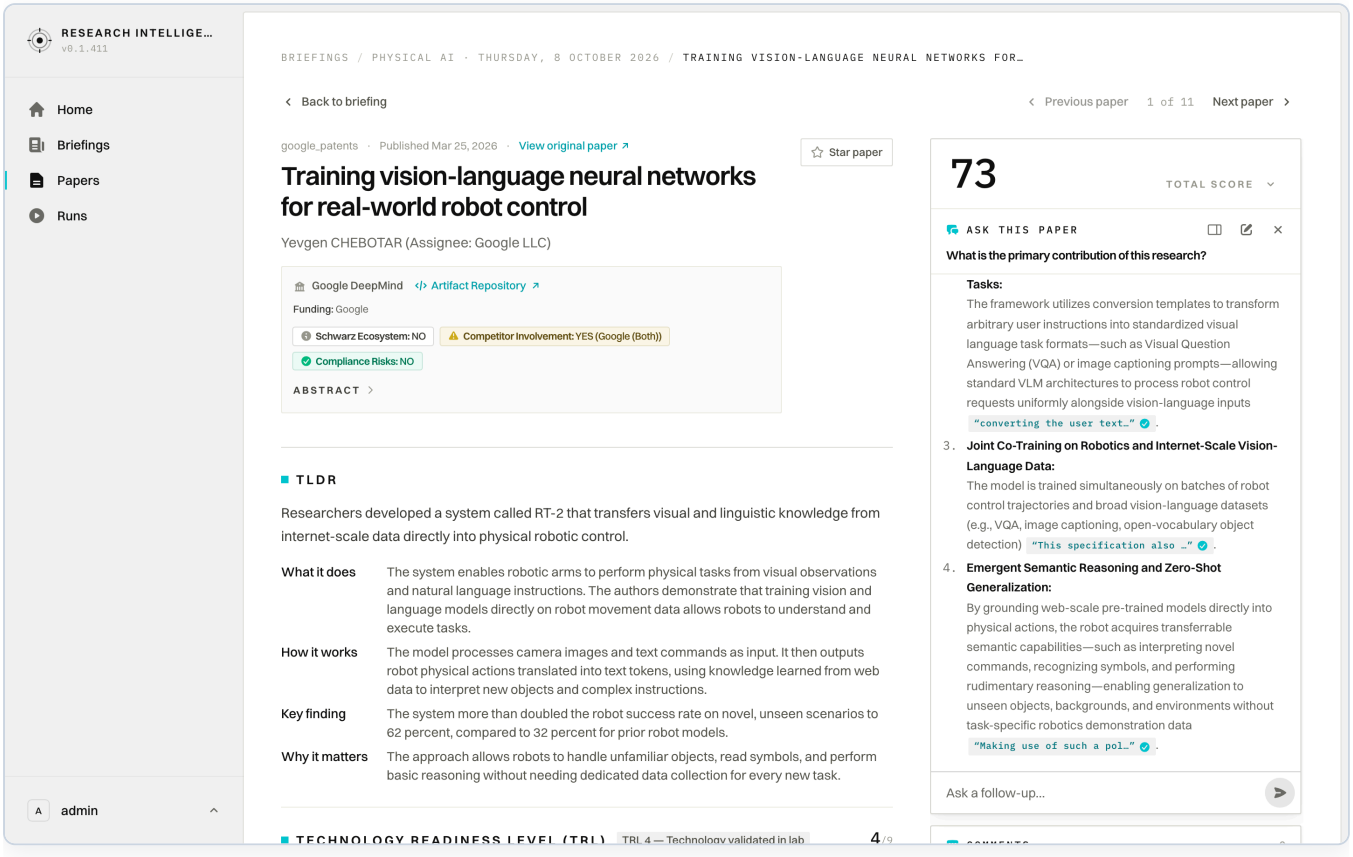
# User Journey: Ask Document Interactive Research Assistant

## 1. Overview & Objective

The **Ask Document Journey** provides an on-demand AI interrogation console that enables researchers to converse directly with an individual scientific publication. Rather than answering from broad general knowledge, the assistant grounds its answers strictly in the paper's full-text PDF, delivering cited excerpts, architectural breakdowns, and methodology critiques without hallucination.

## 2. Interface View

The Ask Document drawer slides out over the right side of the paper view, maintaining full context of the underlying publication:




## 3. Navigation & Interaction Triggers

- **Primary Entry Point**: Located on any Paper Detail page ( `/papers/[id]` ).

- **Trigger Actions**:
  1. **Rail Composer**: Typing a query into the *"What would you like to know?"* input beneath the score rail and pressing `Enter`.
  2. **Mobile Floating Pill**: On smaller viewports where the right rail collapses, a floating action button (`Ask about this paper`) launches the drawer.
- **Drawer Controls**:
  - `×` (Top-right close button) or pressing `Escape` closes the drawer and ends the active interrogation session.

---

## 4. UI Components & Screen Topology

The Ask Document drawer is an animated slide-over panel:

### 4.1 Drawer Header

- **Eyebrow & Icon**: Teal speech-bubble icon with `ASK THIS PAPER`.
- **Context Subtitle**: Full title of the paper under examination (e.g., *"RushHour: A Dynamically Reconfigurable Lattice-Surgery Architecture"*).
- **Dismissal Button (`×`)**: Closes the drawer and cleans up active stream listeners.

### 4.2 Conversation Transcript

- **User Inquiries**: Highlighted prompt bubbles displaying user questions (e.g., *"What is the primary contribution of RushHour architecture?"*).
- **Live Streaming Indicator**:
  - Teal pulsing status dot with live phase indicator: `Reading the full text... 5.0s`.
  - Words stream progressively into view as the inference engine synthesizes the answer.
- **Visual Citations**: Inline markers (`[Page N, Section X]`) linking directly to specific passages in the paper's layout markdown or PDF coordinates.
- **Answer Presentation**: Rich Markdown formatting with syntax highlighting, bulleted technical lists, and formula notation.

### 4.3 Drawer Footer Composer

- **Input Field**: Persistent query input (`Ask a question about this paper...`).
- **Send Trigger**: High-contrast circular action button with paper airplane icon (`➤`).
- **State Handling**: Input automatically locks during live generation to maintain conversational turn ordering.

---

# 5. Step-by-Step User Workflow

## 5.1 Initiating an Inquiry from Paper Details

1. While reading a paper on `/papers/[id]`, locate the **`ASK THIS PAPER`** composer in the right-hand scorecard column.
2. Enter a specific research question, such as:
   - *"What are the benchmark results compared to existing compilers?"*
   - *"How does the Lattice Management Unit handle ancilla space reconfiguration?"*
3. Press **`Enter`** or click `➤`.
4. The drawer smoothly slides open from the right edge, instantly displaying your question and beginning full-text retrieval.

## 5.2 Engaging in Multi-Turn Technical Dialog

1. Observe the live stream as the assistant reads the full text and synthesizes the findings.
2. Review cited page numbers and section references.
3. Submit follow-up questions in the drawer footer to clarify architectural details or implementation constraints.

## 5.3 Dismissing the Drawer

1. When finished, press `Escape` or click the `×` icon in the top right.
2. The drawer slides away, returning focus immediately to the paper's main abstract and scorecard.

---

# 6. Architectural Rules & Guardrails

- **Strict Full-Text Grounding**: The interrogation agent is explicitly constrained to the active document. It refuses to answer questions unrelated to the paper's domain or content.
- **Ephemeral Sittings**: Each drawer opening represents a focused "sitting". Closing the drawer clears the transcript and drops the session state, ensuring a clean slate for the next reading session.
- **Latency & Streaming Resilience**: Real-time responses are streamed via server-sent events (`POST /bff/chat/stream`). If a network interruption occurs, partial answers are retained alongside a graceful retry notice.

# User Journey: Paper Comments & Annotator Collaboration

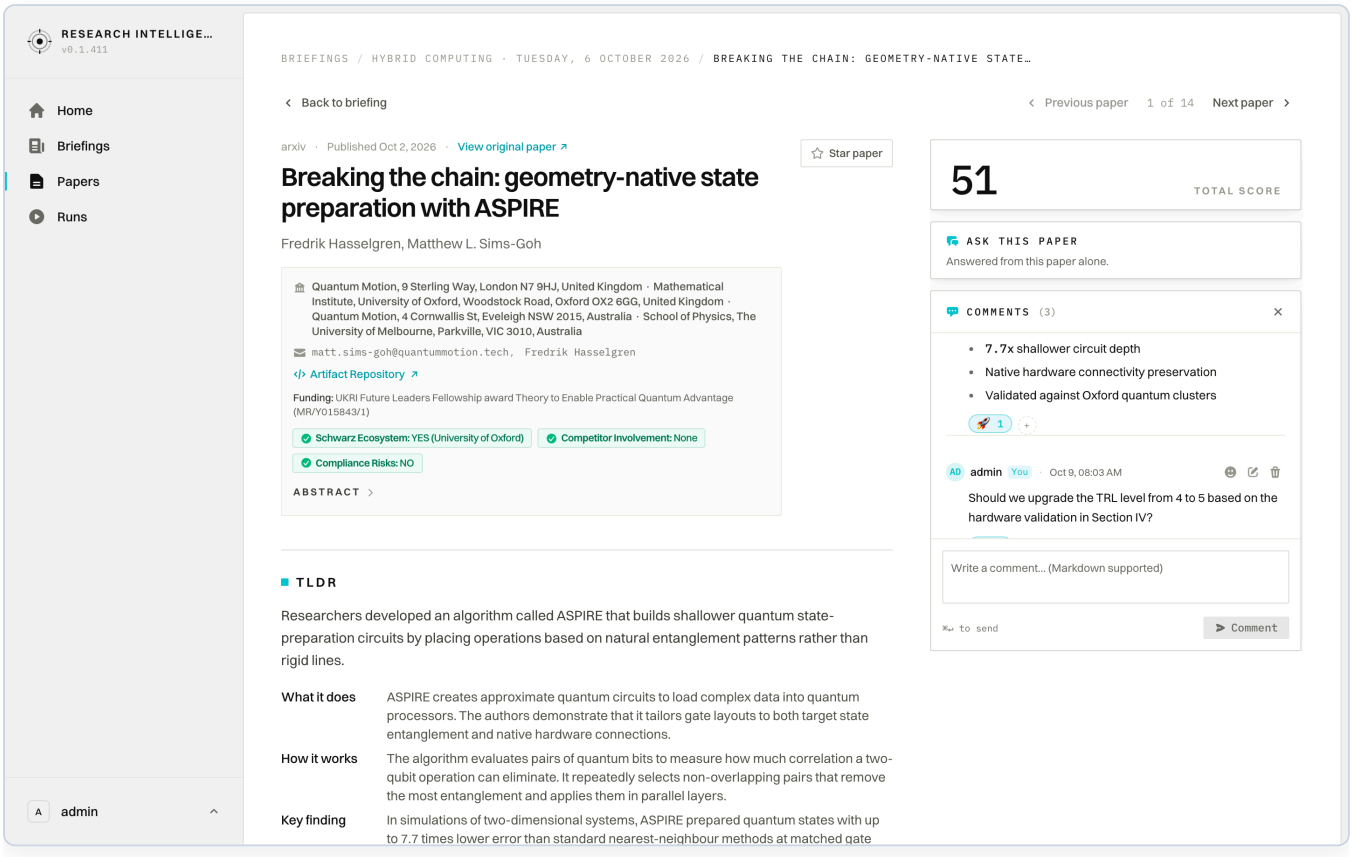
## 1. Overview & Objective

The **Paper Comments & Annotator Collaboration Journey** provides a real-time, threaded workspace console enabling domain specialists, research analysts, and annotators to capture scientific observations, debate evaluation criteria, and establish organizational consensus on high-impact publications.

Rather than isolating evaluation inside static scoring sliders, the comments subsystem integrates an active collaborative layer directly into the paper's review rail. Team members can annotate methodology strengths, question technology readiness (TRL) justifications, format technical findings using GitHub Flavored Markdown, and leverage quick emoji reactions to register agreement without conversational clutter.

## 2. Interface View

The Paper Comments console expands directly beneath the primary score rail on `/papers/[id]`:



## 3. Navigation & Interaction Triggers

- **Primary Route**: Paper Detail View ( `/papers/[id]` ).
- **Access State**:
  - **Collapsed Rail Card**: When collapsed, the card occupies minimal vertical height, displaying the total comment count (e.g., `COMMENTS (3)` ) and an invitation prompt ( `Add a comment…` ).
  - **Expanded Console**: Clicking anywhere on the card header or prompt smoothly opens the interactive discussion feed and composer.
- **Console Dismissal**: Clicking the `×` (Collapse button) returns the rail to its compact footprint while preserving all saved comment state.

---

## 4. UI Components & Screen Topology

The Paper Comments interface is architected for readability, rapid consensus, and precise scientific exchange:

### 4.1 Console Header

- **Typographic Identity**: Distinctive teal speech-bubble icon with uppercase tracking: `COMMENTS` .
- **Live Counter**: Monospace item counter (e.g., `(3)` ) reflecting the active message volume.
- **Collapse Action**: Quick `×` button to tuck the console away when focusing purely on document prose or the Ask Document assistant.

### 4.2 Interactive Message Feed

- **Author Identity**: Two-letter avatar badge derived from author credentials, accompanied by full username or display name.
- **Self Badge**: An explicit `You` pill highlights messages authored by the active logged-in user.
- **Timestamp & Audit Trail**: Relative and formatted ISO timestamp, accompanied by an `(edited)` indicator whenever a comment draft has been updated.
- **Rich Markdown Formatting**:
  - Unordered bullet lists ( `*` , `-` ) and numbered steps ( `1.` , `2.` ).
  - Technical emphasis ( `**bold**` , `*italic*` , `~~strikethrough~~` ).
  - Hard line breaks for natural readability.
  - Monospace code blocks and formulas.

### 4.3 Multi-Action Header & Management

- **Reaction Trigger ( `FaceSmileIcon` )**: Opens an on-demand emoji picker palette with 6 standardized scientific review reactions:
  - 👍 `thumbs_up` : Agree / Methodologically Solid

- 👎 `thumbs_down`: Disagree / Concerns Raised
- 💡 `bulb`: Insightful Observation
- 🚀 `rocket`: Priority / High-Impact Breakthrough
- ❓ `question`: Clarification Needed
- ❤️ `heart`: Appreciation / Commendation

- **Inline Editing (`PencilSquareIcon`)**: Author-exclusive quick editing mode with `esc` to cancel and `⌘↵` to save changes.
- **Audited Deletion (`TrashIcon`)**: Removes comments with a modal confirmation safeguard (`Delete comment? This action cannot be undone`).

### 4.4 Reaction Chips & Consensus Bar

- **Pill Counters**: Reactions render directly beneath each comment as clickable rounded chips (e.g., `[👍 2]`, `[🚀 1]`).
- **User Tooltips**: Hovering over any reaction chip displays the names of contributing team members.
- **Instant Toggle**: Clicking an active reaction chip toggles the user's vote instantly with optimistic UI feedback.
- **Quick `+` Chip**: Allows colleagues to append additional reaction types directly from the reaction bar.

### 4.5 Markdown Composer

- **Input Area**: Multi-line auto-resizing text area with helpful placeholder: `Write a comment… (Markdown supported)`.
- **Keyboard Shortcut**: `⌘↵` (macOS) or `Ctrl+Enter` (Windows/Linux) triggers instant comment submission.
- **Auto-Scroll Behavior**: Upon opening or posting, the message scroller automatically scrolls to the latest message, ensuring researchers never miss ongoing discussion threads.

---

## 5. Step-by-Step User Workflow

### 5.1 Reading Existing Team Observations

1. Navigate to a surfaced document on `/papers/[id]`.
2. Review the right-hand column beneath the score card to see if comments have been logged.
3. Click the `COMMENTS` header to expand the full discussion transcript.
4. Read through observations left by other domain specialists.

### 5.2 Posting an Annotator Observation

1. In the expanded console, click into the composer textarea.
2. Structure your note using markdown (e.g., listing key verification points or code availability).
3. Press `⌘↵` or click `➤ Comment`.
4. The message appears instantly at the bottom of the feed with your avatar and name.

### 5.3 Building Team Consensus with Reactions

1. Identify a comment proposing a score calibration or highlighting a breakthrough.
2. Click the `😀` icon or the `+` pill to open the reaction palette.
3. Select an emoji (such as `👍` to endorse the finding or `🚀` to flag organizational priority).
4. Observe the counter increment, signaling alignment across the intelligence team.

---

## 6. Access Control & Collaboration Governance

- **Cross-Domain Isolation**: Comments strictly adhere to multi-tenant workspace boundaries; observations on proprietary or pre-publication literature are never exposed across foreign domains.
- **Attribution & Edit Tracking**: Every modification updates the audit timestamp and records an `is_edited` flag, maintaining scientific integrity for audit logs.
- **Staff Moderation**: Workspace administrators possess administrative rights to remove non-compliant annotations, while regular analysts manage their own contributions.

# User Journey: Cross-Domain Corpus & All Papers Directory

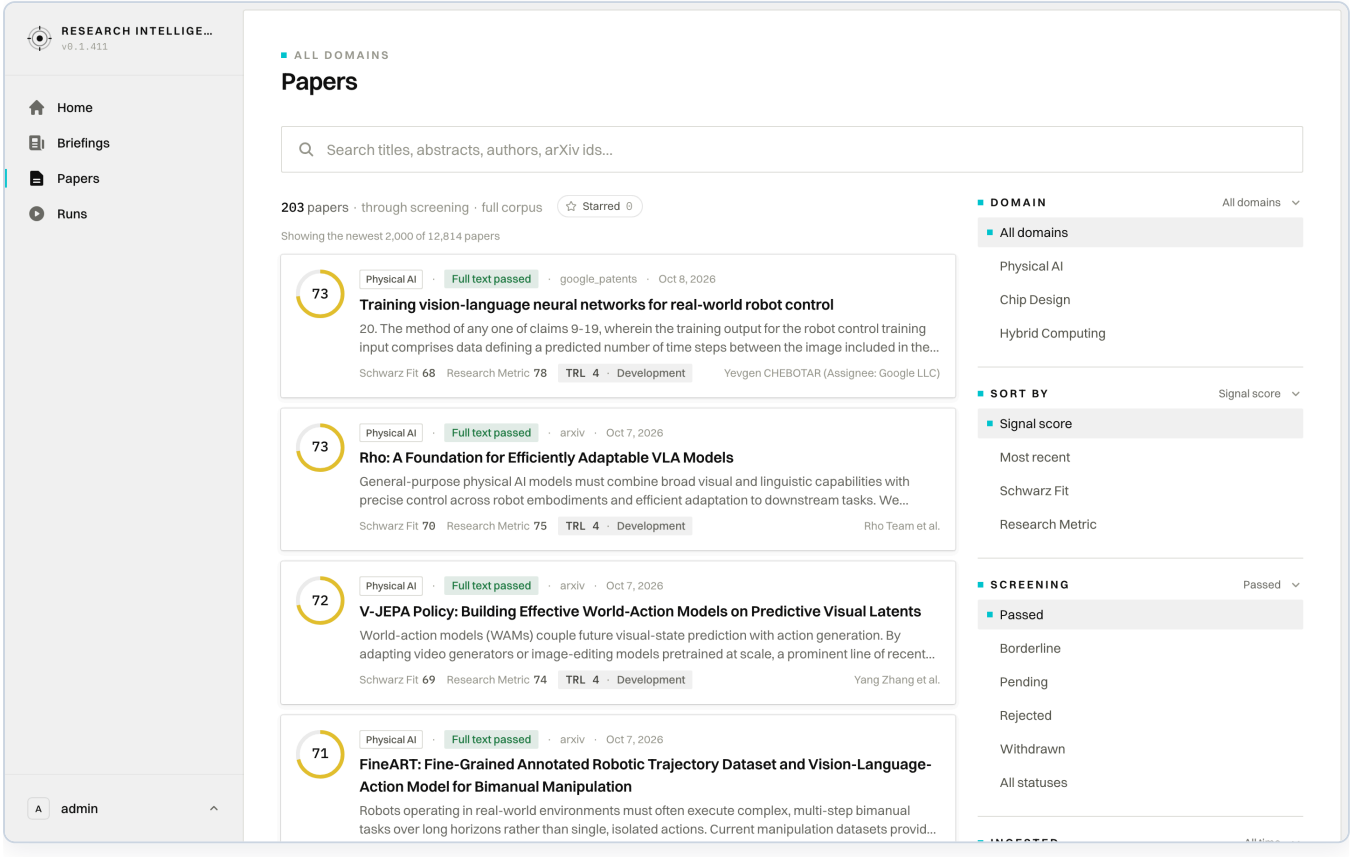
## 1. Overview & Objective

The **Cross-Domain Corpus & All Papers Directory Journey** provides a unified catalog browser over all scientific literature ingested, screened, and analyzed across all enterprise research domains.

While briefings organize literature into curated daily or ad-hoc runs, the `/papers` directory serves as the definitive reference library. Analysts and domain leads can search across thousands of indexed documents, filter by enterprise domains (e.g., Physical AI, Retail Automation, Cyber Security), inspect screening pass yields, isolate starred benchmark controls, and explore literature independently of specific execution dates.

## 2. Interface View

The `/papers` catalog interface displays a comprehensive facet-filtered paper registry:



## 3. Navigation & Filtering Mechanics

- **Primary Route**: `/papers` (Accessible via sidebar navigation).
- **URL-Preserved State Architecture**: The entire filter state is preserved in the URL query string, making search filters and result pages shareable:
  - `?domain=[id]`: Narrows the catalog to a single strategic research domain.
  - `?q=[term]`: Full-text search matching titles, abstracts, author names, and identifiers.
  - `?starred=1`: Filters strictly for items marked with the golden star benchmark flag.
  - `?sort=score` / `?sort=date`: Sorts by total composite score or publication recency.
  - `?page=[n]`: Paginates across the multi-thousand document corpus.

---

## 4. UI Components & Screen Topology

The All Papers catalog layout provides high-density, scannable research intelligence:

### 4.1 Page Header & Eyebrow

- **Eyebrow Context**: Displays `ALL DOMAINS` or the specific active workspace name.
- **Main Heading**: Clean, prominent `Papers` masthead.
- **Corpus Counter**: Displays total indexed volume passing through automated screening (e.g., `203 papers · through screening · full corpus`).

### 4.2 Facet Filter Toolbar

- **Domain Selector Tabs**: Quick horizontal chip buttons allowing researchers to jump between domains or examine the cross-domain aggregate.
- **Starred Toggle Chip**: Quick filter showing current starred paper volume ( `Starred · 0` ) to quickly isolate flagged benchmark papers.
- **Global Search Input**: Interactive search bar with instant query debounce for locating specific authors, titles, or concepts.
- **Dynamic Sort Dropdown**: Toggles between highest signal score first, publication date, or individual evaluation criteria.

### 4.3 Paper Card Grid / List

Each paper card in the directory surfaces actionable intelligence without requiring a click-through:

- **Composite Score Badge**: Monospace total score figure (e.g., `73`, `51`) in high-contrast ink.
- **Domain & Pipeline Badge**: Domain attribution chip (e.g., `Physical AI`), source pipeline tag ( `Full text passed` ), and scraper provenance ( `google_patents`, `arxiv` ).
- **Publication Date**: Formatted publication date (e.g., `Oct 8, 2026`).

- **Paper Title & Link**: High-contrast headline linking directly to the full `/papers/[id]` detail view.
- **Author Summary**: Academic attribution (`et al.` formatted).
- **Executive Abstract / TLDR Excerpt**: Two-line synthesized summary outlining what the paper enables and how it works.
- **Interactive Star Control**: Instant toggle to bookmark or promote papers to team benchmark status.

### 4.4 Pagination & Range Footer

- **Range Summary**: Clear indicator of current page window (e.g., `Showing the newest 2,000 of 12,814 papers`).
- **Pager Navigation**: Numeric pagination controls (`Previous`, `Next`, page buttons) ensuring seamless browsing over expansive corpora.

---

## 5. Step-by-Step User Workflow

### 5.1 Exploring the Cross-Domain Corpus

1. Click **`Papers`** in the left application sidebar.
2. Observe the aggregate catalog displaying all literature that successfully passed screening across all active domains.
3. Review the top-scoring papers in the ledger.

### 5.2 Filtering by Domain & Keyword Search

1. Click on a specific domain chip (e.g., `Cyber Security` or `Physical AI`).
2. The ledger dynamically updates to show only literature relevant to that business unit.
3. In the search box, enter a technical keyword (e.g., *"lattice surgery"* or *"vision-language"*).
4. Review matching publications sorted by composite signal score.

### 5.3 Drilling into a Selected Document

1. Identify a high-priority paper in the search results.
2. Click the paper title to transition seamlessly into the **Selected Document Deep-Dive** (`/papers/[id]`).
3. Leverage the scorecard, radar breakdown, Ask Document assistant, and Comments console.

# User Personas & Access Control Specification

**Document Version**: 3.4
**Associated Documents**:

- 🗺️ **user_journeys.md** — Frontend User Journeys & Interaction Sequences
- 📐 **wireframes.md** — Screen Layout Wireframes & UI Component Topologies
- 🧭 **route_matrix.md** — URL Route & Context Resolution Matrix
  **Scope**: Definitions, behavioral workflows, and permission boundaries for the core personas in the Schwarz Research Intelligence (SRI) platform.

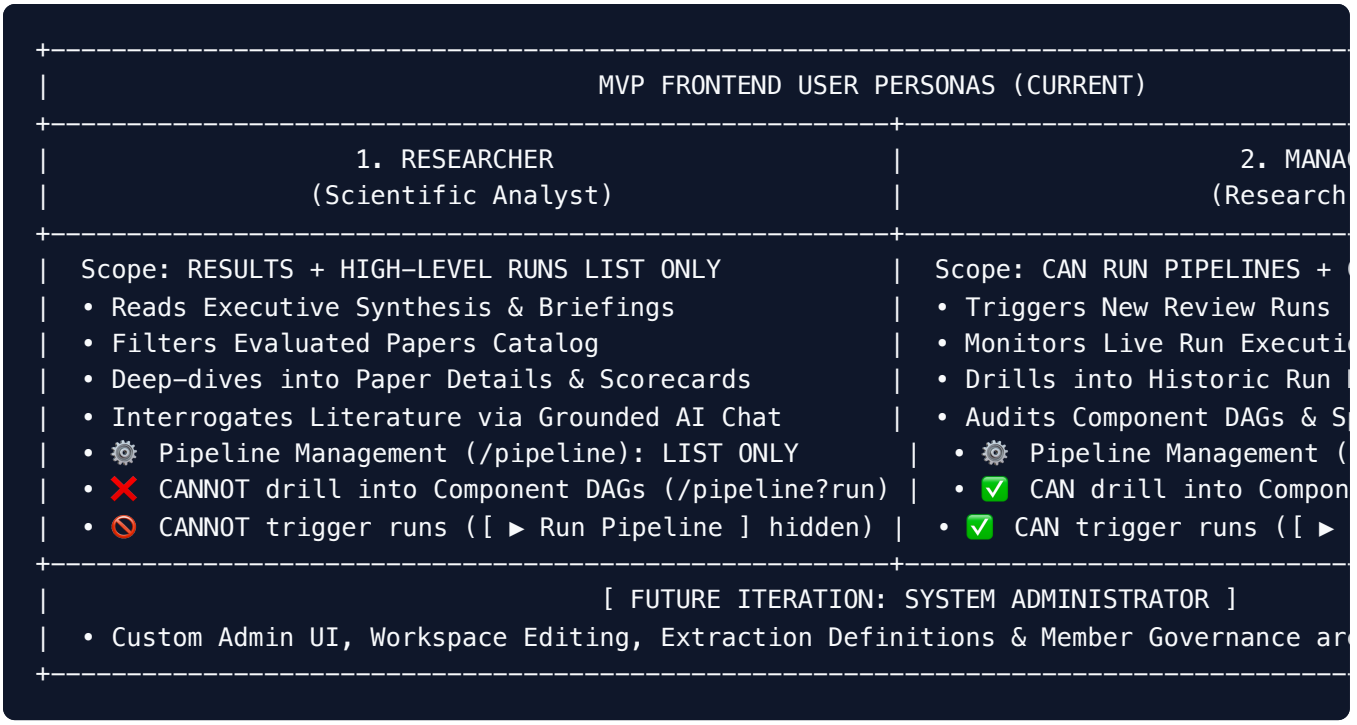
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## 1. Executive Summary: MVP Scope vs. Future Iteration

The frontend MVP is specifically designed around **two active user personas**:

1. **Manager (Research Lead)**: Full operational control over review cycles; can trigger runs ( `[ ▶ Run Pipeline ]` ), drills into **Historic Run Details (Components level)** ( `/pipeline?run_id=...` ), audits DAG execution, and consumes all research outputs.
2. **Researcher (Scientific Analyst)**: Consumes research results ( `/briefing` , `/papers` , `/papers/[id]` , `/ask` ) and has access to **Pipeline Management ( `/pipeline` ) in high-level read-only mode only** (inspects the list of historical runs, completion timestamps, paper counts, and durations). **The Researcher CANNOT trigger runs and CANNOT drill into component-level DAG execution details.**

> IMPORTANT
>
> **Admin Features are Planned for Future Iteration**:
> In the current MVP phase, **we do not have anything specific for Admin on the frontend**. System administration, multi-tenant workspace provisioning, scraper query configuration, and fine-grained quota governance are reserved for a **Future Iteration**. Any underlying maintenance in the current phase is handled via standard backend interfaces without custom frontend screens.




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## 2. Persona 1: Researcher (Scientific Analyst)

### Profile & Motivation

- **Role Title**: Senior Research Scientist, Domain Analyst, Postdoctoral Researcher, Technology Scout.
- **Primary Goal**: Quickly discover breakthrough papers in their domain, understand why a paper received a high Signal Score, verify technical claims against full-text evidence, interrogate methodologies using AI, and maintain awareness of when review runs took place.
- **Daily Pain Points Solved**:
  - Eliminates manual screening of hundreds of noisy arXiv / OpenAlex preprints.
  - Replaces manual PDF scanning with structured scoring ledgers, radar charts, and LLM rationales.
  - Enables instant natural-language Q&A directly cited from paper sections and figures.
  - Provides a clean overview of when historical SLR review cycles were executed.

### Access Boundary: Results + High-Level Runs List Only

The Researcher has full visibility into research outputs and a high-level summary of review runs, but **is strictly barred from technical component-level pipeline details and operational triggers**:

- **Accessible Views**:
  1. 📊 **Briefings Dashboard (`/briefing`)**: Reads executive summaries, breakthroughs, bottleneck analysis, and top-scoring paper cards.

2. 📄 **Browse Papers Catalog (`/papers`)**: Filters papers by screening status, signal score slider, TRL window, and free-text search.
3. 📑 **Paper Details & Scorecard (`/papers/[id]`)**: Deep-dives into AST formula ledgers, radar charts, LLM screening reasons, and canonical PDF files.
4. 💬 **Ask Paper Console (`/ask`) & In-Page Assistant**: Converses with papers, verifies citation badges, and restores chat history threads.
5. ⚙️ **Pipeline Management & Runs Audit (`/pipeline`) — READ-ONLY RUNS LIST ONLY**:
   - **Visible**: Inspects high-level list of historical runs, completion timestamps, paper counts, and duration metrics.
6. 👤 **Personal Account Settings (`/settings`)**: Updates name, email, and password.

- **Strictly Restricted Views & Actions**:
  - ❌ **CANNOT Drill into Component Details**: **Historic Run Details at the component level (`/pipeline?run_id=...`) is strictly HIDDEN from Researchers**. Researchers cannot click run rows to open the component DAG breakdown, cannot view individual stage logs (Search, Download, OCR, Screening, Extraction, Signal, Report), and cannot view internal component execution latencies.
  - 🚫 **CANNOT Trigger Runs**: The `[ ▶ Run Pipeline ]` CTA button is **hidden or disabled** for Researchers (*"Manager role required to initiate pipeline runs"*).
  - 🚫 **CANNOT Abort or Modify Runs**: Only Managers have operational pipeline authority.

### Typical User Journey

1. **Morning Scan**: Opens `/briefing` for the latest successful run to review the executive synthesis and top 3 high-signal papers.
2. **Targeted Investigation**: Clicks a paper card to open `/papers/[id]?run_id=...` and inspects the Signal Score AST formula breakdown and TRL radar chart.
3. **In-Page Interrogation**: Uses the integrated `Ask this paper` right rail to ask: *"Explain the readout latency assumptions in Section 3.2"*.
4. **Run Awareness**: Navigates to `/pipeline` to check when the last review cycle completed, viewing its completion timestamp and paper count in the summary table.

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## 3. Persona 2: Manager (Research Lead / Project Director)

### Profile & Motivation

- **Role Title**: Research Director, Lab Head, R&D Program Manager, Innovation Lead.
- **Primary Goal**: Track research velocity, evaluate domain maturity (TRL progression), initiate new automated review cycles, and perform in-depth component-level audits of Systematic Literature Review (SLR) execution quality and spend.

- **Daily Pain Points Solved**:
  - Provides instant high-level visibility into review funnel yields (screened vs. passed).
  - Enables on-demand triggering of review runs without risking duplicate concurrent jobs.
  - Provides detailed component-level execution transparency (Search ➔ Ingestion ➔ Screening ➔ Extraction ➔ Scoring ➔ Reporting) to diagnose stage bottlenecks and audit spend.

### Access Boundary: Can Run Pipelines + Component-Level DAG Audit + Results

The Manager has complete access to all research outputs, full operational control to trigger runs, and exclusive access to drill into component-level execution details.

- **Accessible Views**:
  1. **All Researcher Views**: Full access to `/briefing`, `/papers`, `/papers/[id]`, `/ask`, and high-level `/pipeline`.
  2. ⚙️ **Pipeline Management (`/pipeline`) — FULL OPERATIONAL CONTROL**:
     - **Trigger Pipeline Runs**: **Can start a fresh review cycle using the `[ ▶ Run Pipeline ]` button** (protected by a single active run concurrency mutex lock).
     - **Live Execution Monitoring**: Observes run status in real time while stages progress.
     - **Historic Runs Overview**: Reviews past runs table showing timestamps, durations, paper yields, and total USD costs.
     - **Exclusive Component DAG Audit (`/pipeline?run_id=...`)**: **Drills into Historic Run Details (Components level)** by clicking any run row. Audits individual stage outputs (Search ➔ Ingestion ➔ Screening ➔ Extraction ➔ Scoring ➔ Reporting), inspects stage timings, models, and component costs in read-only mode (`No Re-run`).
  3. 🔔 **Notification Preferences (`/settings`)**: Configures email digest alerts upon run completion.

### Typical User Journey

1. **Executive Review**: Reviews `/briefing` on Monday morning to evaluate the weekly review output.
2. **Funnel Audit**: Checks `/pipeline` to ensure Run #14 completed within budget ($1.15) and processed all fetched papers (142 fetched, 18 passed).
3. **Component Deep Dive**: Clicks Run #14 on `/pipeline` to open the **Historic Run Details (Components level)** drawer and verifies the screening and extraction stage latencies.
4. **Initiating Reviews**: If new papers were preprinted after a major conference, clicks `[ ▶ Run Pipeline ]` on `/pipeline` to trigger an on-demand review cycle.

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# 4. Persona 3: System Administrator (Future Iteration)

## MVP Scope Status

> **NOTE**
>
> **No Dedicated Admin UI in MVP**:
> For the initial MVP, there are no dedicated frontend screens or user journeys for the System Administrator. Governance and administrative capabilities are planned for a **Future Iteration**.

## Future Iteration

In a future iteration, a dedicated administration portal will be introduced encompassing four core capabilities:

1. **Workspace Management & Editing**: Creating, editing, configuring, and archiving research domain workspaces and their global attributes.
2. **Extraction Definitions**: Defining, customizing, and editing domain-specific extraction parameters, schema rules, and AST scoring variables.
3. **Workspace Members & Roles**: Provisioning user accounts, managing workspace team memberships, invitations, and role assignments (`manager` vs. `researcher`).
4. **Upstream Scrapers & Quotas**: Configuring search queries, data sources (arXiv, OpenAlex, PubMed), scraping rate limits, and monthly token/USD budget ceilings.

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# 5. Screen & Action Permissions Matrix (Current MVP)

The following matrix defines the exact visibility and action capabilities across the active MVP personas:

| View / Action | URL Route | Researcher (Analyst) | Manager (Lead) | System Admin (Future) | Notes |
|---|---|---|---|---|---|
| **Login Screen** | `/` | ✅ Visible | ✅ Visible | *(Future)* | Standard authentication for all active roles. |
| **Briefings Dashboard** | `/briefing` | ✅ Visible | ✅ Visible | *(Future)* | High-level synthesis and top paper cards. |
| **Browse Papers Catalog** | `/papers` | ✅ Visible | ✅ Visible | *(Future)* | Multi-attribute search, TRL & score filters. |
| **Paper Details & Scorecard** | `/papers/[id]` | ✅ Visible | ✅ Visible | *(Future)* | AST formula ledger, radar chart, PDF link. |
| **Ask Paper Console** | `/ask` | ✅ Visible | ✅ Visible | *(Future)* | Grounded multi-turn conversational chat. |
| **In-Page 'Ask this paper'** | `/papers/[id]` rail | ✅ Visible | ✅ Visible | *(Future)* | Contextual document assistant. |
| **Pipeline Runs Overview (High-level)** | `/pipeline` | ✅ **Visible (List Only)** | ✅ **Visible (Active)** | *(Future)* | **Researcher sees list of runs only.** |
| **Trigger Pipeline Run** | `/pipeline` CTA | 🚫 **No (Manager Only)** | ✅ **Can Run (`[ ▶ ]`)** | *(Future)* | Subject to single active run mutex lock. |
| **Historic Run Details (Components level)** | `/pipeline?run_id=...` | 🚫 **Hidden (Manager Only)** | ✅ **Visible & Active** | *(Future)* | **Researcher cannot enter into component details.** |
| **Audit Spend & Run Metrics** | `/pipeline` | ✅ **Visible (Table summary)** | ✅ **Visible (Component DAG)** | *(Future)* | Researcher sees summary table costs; Manager audits stage spend. |
| **Manage Profile &** | `/settings` | ✅ Visible | ✅ Visible | *(Future)* | Self-service credential |

| View / Action | URL Route | Researcher (Analyst) | Manager (Lead) | System Admin (Future) | Notes |
|---|---|---|---|---|---|
| **Password** | | | | | update. |
| **Workspace Management & Editing** | Dedicated Admin UI | 🚫 No | 🚫 No | ⏳ **Future Iteration** | Out of scope for MVP frontend. |
| **Extraction Definitions** | Dedicated Admin UI | 🚫 No | 🚫 No | ⏳ **Future Iteration** | Out of scope for MVP frontend. |
| **Workspace Members & Roles** | Dedicated Admin UI | 🚫 No | 🚫 No | ⏳ **Future Iteration** | Out of scope for MVP frontend. |
| **Upstream Scrapers & Quotas** | Dedicated Admin UI | 🚫 No | 🚫 No | ⏳ **Future Iteration** | Out of scope for MVP frontend. |

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*End of User Personas & Access Control Specification.*